



Financial expert CEOs: CEO's work experience and firm's financial policies[☆]



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ABSTRACT

We study CEOs with a career background in finance. Firms with *financial expert CEOs* hold less cash, more debt, and engage in more share repurchases. Financial expert CEOs are more financially sophisticated: they are less likely to use one companywide discount rate instead of a project-specific one, they manage financial policies more actively, and their firm investments are less sensitive to cash flows. Financial expert CEOs are able to raise external funds even when credit conditions are tight, and they were more responsive to the dividend and capital gains tax cuts in 2003. Analyzing CEO-firm matching based on financial experience, we find that financial expert CEOs tend to be hired by more mature firms. Our results are consistent with employment histories of CEOs being relevant for corporate policies. However, we cannot formally rule out that our findings are partly explained by endogenous CEO-firm matching.

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1. Introduction

There is evidence that chief executive officers (CEOs) have particular management styles and that they matter for corporate performance (see Adams, Almeida, and Ferreira, 2005; Bennedsen, Perez-Gonzalez, and Wolfenzon, 2008). Most evidence on CEO-specific heterogeneity examines education, personal characteristics, or personality traits. Malmendier and Tate (2005) study education, Kaplan, Klebanov, and Sorensen (2012) personal characteristics, and Malmendier and Tate (2005, 2008), Graham, Harvey, and Puri (2013), Hirshleifer, Low, and Teoh (2012), Malmendier, Tate, and Yan (2011), and Lin, Ma, Officer, and Zou (2011) analyze personal traits. Less is known about the work experience of CEOs. An exception is Custódio and Metzger (2013) who study CEOs' industry expertise in the context of diversifying mergers and acquisitions (M&A) activity.

We contribute to the literature on CEO style in three ways. First, we show that previous financial expertise of CEOs is correlated with firms' financial policies. Second, we provide micro-level evidence on how financial experience operates. Third, we explicitly discuss the possibility that endogenous matching between CEOs and firms based on *time-varying characteristics* biases our results. Even though it is very difficult in general to establish a clean causal link between financial expertise and firm policies because exogenous variation in the financial expertise of the CEO is simply not observed, we provide a coherent and vast set of results that support the view that financial expertise matters. We also make use of both exogenous variations in the business conditions in which the CEO operates and CEO turnovers that are likely to be exogenous.

We first show that an important fraction of CEOs are *financial expert CEOs*. That is, 41% of the CEOs have previous experience in the financial industry or in a financial role. When we look at their firm's financial policies, we find that nonfinancial firms headed by financial expert CEOs hold less cash on average. Furthermore, they are more leveraged and they have a higher propensity than other firms to pay out money to shareholders. These differences are economically meaningful. Regression results indicate that cash holdings are about 12% lower and average leverage ratios are 6% higher (evaluated at the mean). These firms are also 7% more likely to repurchase shares.

Evaluating financial expert CEOs' actions and decisions at a micro-level, we provide evidence that they are more financially sophisticated. Typical firms use companywide discount rates to evaluate investment projects rather than a project-specific one. This has been called the weighted average cost of capital (WACC) fallacy (Graham and Harvey, 2001; Krüger, Landier, and Thesmar, 2011). Following Krüger, Landier, and Thesmar (2011), we use segment-level investment policy in diversified firms to find that only firms without a financial expert CEO fall into the WACC fallacy trap. We then exploit exogenous changes to business conditions to test whether financial experts react in a way that is consistent with higher levels of financial sophistication. We use the 2003 cuts to dividend and capital gains taxes, known as the "*Bush tax cuts*," and find that financial experts were more responsive as measured by increases in total payout to shareholders. The tax cuts increased the dollar value of dividends and capital gains after 2003, therefore, increasing payout would be more beneficial to shareholders. We also show that financial expert CEOs are better able to raise external financing, even under tighter credit market conditions, suggesting that they have better access to capital markets. These results are consistent with the idea that financial experts can follow more aggressive financial policies (holding less cash and more debt) because they can access financial markets more easily.

Guner, Malmendier, and Tate (2008) show in their research on board of directors' composition that financial experts have better access to capital markets. They show that firms with bankers on their boards use more external financing and have lower investment-cash flow sensitivity.

However, the presence of bankers on a board is associated with policies that tend to benefit those financial institutions, but not necessarily the lending firms' shareholders. In fact, banks seem to provide funding to firms with good credit but poor investment opportunities.¹ We also show lower investment-cash flow sensitivity for financial expert CEOs, but interestingly, the benefits of CEO financial expertise do not come at the cost of benefiting financial institutions. We find no evidence suggesting that financial experts overinvest just because they have better access to financing. This result seems plausible, as CEOs with financial expertise do not have a conflict of interest as bankers on the board do.

Additional evidence that financial expertise matters is that newly hired financial expert CEOs are more likely to replace an incumbent chief financial officer (CFO) within the first year of their hire. We interpret this result as indication that financial expert CEOs are more involved with firm financial policies. It is likely to be easier to implement changes in a firm by changing a management team, in this case the CFO. Of course, we cannot rule out the alternative hypothesis that CFOs resign voluntarily more often if a financial expert CEO is hired.

Financial expert CEOs might also directly benefit from their personal links to the financial sector. We test this hypothesis by examining whether former employers of the CEO are providing loans to his or her firm. There is no evidence that these direct links are driving our results. We use data on syndicated bank loans from Dealscan to show that only a very small fraction (2.1%) of loan facilities is provided by former employers of the CEOs. We also do not find evidence that financial expert CEOs are more likely to use indirect links beyond their previous employers, as they are no more likely to borrow from a new lending institution (a bank that was not providing loans to the company before the CEO was appointed).

Overall, our findings also contribute to the literature that tries to understand cross-sectional variation in cash holdings, leverage, and payout policy. Early survey evidence by Graham and Harvey (2001) and more recently by Lins, Servaes, and Tufano (2010) suggest that the managerial impact on financial policies is likely to be big. They show that managers largely differ in their criterions and application of financial theories when it comes to capital structure.

A common limitation in the literature on CEO characteristics is interpretation of the results. There are at least two ways that selection can bias the estimation of any potential effect of a CEO characteristic such as financial expertise on firm decision making. First, there might be omitted variables on the CEO level. For instance, financial expertise might be correlated with other CEO characteristics such as talent or education that drive our results. We address this concern in several ways. We control for education, general expertise, and talent using detailed biographical data. We also show direct evidence of financial sophistication by financial experts. This increases our

¹ Dittmann, Maug, and Schneider (2010) show similar findings in Germany.

confidence that our measure of financial expertise is not a proxy for something else.

There is also a concern that endogenous CEO-firm matching could bias the results. There might be unobserved firm heterogeneity that simultaneously explains the matching between firms and financial expert CEOs as well as firm financial policies. For instance, financial expert CEOs might have preferences for certain firms that also happen to have low cash and high leverage ratios. Researchers are well aware of this problem and have addressed it by exploiting within-firm variation.² In our analysis, we also control for firm fixed effects.

The fixed effects approach, however, addresses endogenous matching based only on *time-invariant* characteristics. Although most authors do not explicitly address it, this concern is not new. It is recognized in the seminal work of [Bertrand and Schoar \(2003\)](#), which shows that CEO fixed effects are significant across several firm policies. Bertrand and Schoar acknowledge that their results are consistent with two alternative interpretations: (1) CEOs impose their idiosyncratic styles on companies, and (2) boards choose CEOs because of their attributes, in case firms' optimal strategies change over time. One contribution of our work is direct evidence on the matching between firms and CEOs based on both observable CEO and observable firm characteristics.³ We show that financial expert CEOs tend to be appointed by mature firms, while non-experts tend to be appointed by firms in the growth stage of their life cycle. This is consistent with assortative CEO-firm matching based on the life cycle of the firm and the career background of the CEO.⁴ Endogenous CEO-firm matching is not necessarily inconsistent with a causal interpretation of the documented effects of financial policies. Indeed, even though the life cycle of a firm might determine its optimal financial policies, the financial expertise of the CEO might still be crucial for implementing these strategies ("financial skills-based view"). That is, without the financial expertise of the CEO, the firm could not implement these optimal policies. Under an alternative hypothesis, firms are able to implement any financial policy, whatever the identity of the CEO. Thus, the effects observed originate purely from the preferences, or other unobserved characteristics of financial expert CEOs who self-select firms that implement specific financial policies. Under this "no financial skills

view," appointing a financial expert CEO could be the *cheapest* way to implement a given strategy, either because they have a preference to run a firm under those policies, or because for some other reason—other skills for instance—it is easier for the CEO to run the firm in such a way.

For example, assuming that for some reason financial expert CEOs have a preference for running mature firms, companies in that stage of their life cycle might hire them because they might be willing to work at a lower salary. We directly test whether compensation for financial expert CEOs and non-experts is different in level or structure. We do not find any differences. Consistent with the notion that financial skills are crucial and valuable (in contrast to a "no financial skills" view), we also show that there is a positive valuation effect of financial expert CEOs in mature companies.

We argue that our findings are consistent mostly with a financial skills-based interpretation of the estimated effects of financial expertise, even though we cannot formally reject the alternative view. As already noted, we provide direct evidence that financial experts have skills and are financially more sophisticated in application of finance theory. Moreover, exploiting exogenous variation in the business environment, we find that financial expert CEOs are able to access credit markets in unexpectedly tight conditions and react quickly to personal tax changes in a way that is beneficial to shareholders.

We intend to control for endogenous matching based on the life cycle of the firms by directly including further firm-level controls and by using a matching estimator. We also analyze likely exogenous timing of a turnover. We look at a subsample of firms replacing a retiring CEO and the results hold in this subsample. Even though we are aware that the choice of the new CEO is endogenous, the timing of the turnover is likely exogenous, and this can help us to identify the effect of financial expertise when these firms switch from a *nonfinancial expert CEO* to a *financial expert CEO*.

Finally, we analyze whether financial expertise of the CEO comes at the cost of other skills relevant to the firm. We focus on innovation, marketing, and labor policies. We find that firms headed by financial experts tend to invest less in research and development (R&D) and generate less innovation as measured by the number of filed patents and patent citations. When we analyze advertising expenses, the sensitivity of sales to advertising expenses, and labor costs, we do not find significant differences associated with financial expertise.

The paper proceeds as follows. The next section describes the data. [Section 3](#) analyses firms' financial policies by *financial expert CEOs*. [Section 4](#) provides more micro-evidence on how financial expertise operates. [Section 5](#) discusses alternative interpretations with emphasis on the matching between firms and CEOs. In [Section 6](#) we analyze nonfinancial policies. [Section 7](#) concludes.

2. Sample and data description

This section describes the data sources and presents main summary statistics.

² E.g., [Kaplan, Klebanov, and Sorensen \(2012\)](#) acknowledge endogenous matching as an alternative interpretation; [Malmendier and Tate \(2005, 2008\)](#) and [Malmendier, Tate, and Yan \(2011\)](#) rely on within-firm variation.

³ There is some evidence on endogenous CEO-firm matching that is based on unobservable or time-invariant characteristics. For instance, [Gabaix and Landier \(2008\)](#) and [Tervio \(2008\)](#) study models of assortative matching based on observable firm characteristics (firm size) and unobservable manager characteristics (talent). [Pan \(2012\)](#) studies complementarities between managerial talent and firm size, cross-industry experience of the CEO and diversification, and innovation propensity of the CEO and research and development (R&D) intensity of the firm. [Cronqvist, Makhija, and Yonker \(2012\)](#) show that firms and CEOs match according to their leverage preferences.

⁴ This finding is consistent with the generalized model by [Eisfeldt and Kuhnen \(2013\)](#), who link CEO turnover in firms hit by a negative shock to deterioration of the quality in firm-CEO matches.

2.1. Data

Our initial sample corresponds to the CEO-firm panel from ExecuComp database. This is a sample of 25,562 CEO-firm-years in Standard and Poor (S&P) 1500 firms.⁵ We match the executives identified as CEOs in a given firm-year by ExecuComp with BoardEx to obtain additional CEO characteristics such as education and past financial experience. We perform an additional data match between firms in BoardEx and several other data sets (including Compustat, Datastream, ICARUS, and ORBIS) to obtain the industry classification of past positions for CEOs in the panel and to categorize them as a financial expert or nonfinancial expert.⁶

The sample of past positions of the CEOs includes more than 11,100 firms. Following other researchers, we further exclude from the main CEO-firm panel the financial sector as well as utilities because specific regulations affect their capital structure choice. Our final sample includes 17,716 CEO-firm-years and 4,277 unique CEOs for which we are able to match ExecuComp and BoardEx and for which firm characteristics of interest are available.⁷ The sample period runs from 1993 through 2007. Although ExecuComp provides data since 1992, we drop this year because of the small number of observations (433 CEO-firms).

We obtain accounting data from Compustat, price data from Center for Research in Security Prices (CRSP), analyst forecasts from Institutional Brokers' Estimate System (I/B/E/S) and data on loan facilities from the Dealscan database maintained by the Loan Pricing Corporation.⁸ All variable definitions are reported in Table 1.

2.2. CEO and firm characteristics

Table 2, Panel A, shows descriptive statistics for the financial expertise of CEOs in our panel. A substantial fraction of CEOs (41%) have previous work experience either in the financial industry or in a financial role in a nonfinancial firm. We define a financial expert as a CEO who has past experience in either banking or investment firms (two-digit SIC codes 60, 61, and 62), or in a finance-related role (accountant, CFO, treasurer, or VP of finance), or in a large auditing firm.⁹ Looking at detailed finance experience, we see that 16% of the CEOs have worked in

depository institutions (SIC 60), 5% in non-depository institutions, 6% in investment banks and securities brokerages, and 7% in auditing firms.¹⁰ Panel B shows descriptive statistics for the CEOs in our panel. The CEOs are overwhelmingly male (98%). The typical CEO in our sample is 55 years old, and has been in the company for seven years. Approximately 52% of the panel has some level of financial education, e.g., an MBA, or an economics-related degree (taking into account all levels of education: undergraduate, masters, or doctorate).

Table 2, Panel B shows other average observable characteristics of financial expert and nonfinancial expert CEOs in terms of education and career path. The variables fast track CEO, Ivy League alumnus, and recession graduate are proxies for innate talent, following Falato, Li, and Milbourn (2012) and Custódio, Ferreira, and Matos (2013). Financial expert CEOs differ mainly from nonfinancial experts in their education and previous work experience. Nonfinancial expert CEOs are more likely to have a science degree (44% vs. 24%) or a law degree (21% vs. 9%). Financial experts are more likely to have graduated from an Ivy League school (22% vs. 14%) or in a recession year (31% vs. 27%). We find no significant differences in terms of the age of their first CEO appointment. Financial expert CEOs have more diverse work experience than nonfinancial experts. On average, they have worked for one additional firm and in one additional industry. By the general ability index from Custódio, Ferreira, and Matos (2013) (generalist), we find that financial experts have more general managerial skills (0.21 vs. −0.19). In our regression analysis, we control for these proxies of general skills and talent, as well as age and education. Bertrand and Schoar (2003), and Malmendier and Tate (2005) show that age and having an MBA or other type of financial education are associated with CEO style and with firm financial policies.

Table 2, Panel C shows descriptive statistics for firms in our panel. The average firm has a book value of assets of approximately \$4.4 billion, cash and cash-equivalents that represent 27% of non-cash assets, and book leverage (debt-to-assets) of 22%.

3. Financial expert CEOs and firm financial policy

To examine the relation between firm financial policy and the financial expertise of the CEO, we focus on firm cash holdings, leverage, and payout policy. We run panel regressions on the determinants of cash holdings, leverage, dividend payout, and share repurchases in which the main independent variable of interest is a financial expert CEO dummy.

Columns 1 and 2 of Table 3 show results on whether financial expert CEOs follow different cash holding policies. We run ordinary least squares (OLS) and firm fixed effects regressions in which the dependent variable is the logarithm of cash holdings, i.e., the logarithm of the ratio

⁵ The sample includes both current S&P 1500 and firms that were in S&P 1500 at one time in the sample period but left.

⁶ For the remaining past positions we manually search Standard Industrial Classification (SIC) codes using mainly two Web sites: www.alacra.com and www.manta.com.

⁷ We could not find a match in BoardEx for 1,198 CEOs in our initial panel. The main reason is that BoardEx excludes CEOs who are retired, while ExecuComp does not. This primarily affects the match over the first years of the sample period. Our results are robust to restricting the sample to the most recent period, post 2000, where the ExecuComp-BoardEx match is the greatest (above 90%).

⁸ We use the matching table between Dealscan and Compustat provided by Chava and Roberts (2008).

⁹ We include current and recent top-tier auditing firms such as Pricewaterhouse, Deloitte, Ernst & Young, KPMG, Arthur Andersen, Coopers, Peat Marwick, and Touche Ross. In unreported tests we use alternative definitions of financial expertise, and we also use the individual components of our variable. Our results are robust to the use of these alternative measures.

¹⁰ These figures are similar to those presented by Schoar and Zuo (2011), though there are a few notable differences in terms of past financial experience. These differences are likely to be attributable to a different coverage of the two data sets, but can also be driven by time trends.

Table 1
Variable definitions.

Panel A: CEO characteristics	
Variable	Definition
Board independence	Ratio of number of independent directors to board size.
CEO age	Age of CEO in years.
CEO tenure	Number of years as CEO in the current position.
CEO-chair dummy	Dummy variable equal to one if the CEO is also chairman of the board.
Compensation equity	Equity pay – Value of restricted stock granted plus value of options granted in thousands of dollars (Execucomp RSTKGRNT + + OPTION_AWARDS_BLK_VALUE).(Execucomp).
Compensation total	Total CEO pay in thousands of dollars, which consists of salary, bonus, value of restricted stock granted, value of options granted, long-term incentive payout, and other compensation (Execucomp TDC1). (Execucomp).
Econ dummy	Dummy variable that takes the value of 1 if CEO has an economics-related degree, and zero otherwise.
External hire dummy	Dummy equal to one if the CEO is hired externally; dummy equal to zero if the CEO is internally appointed.
Fast track CEO	Age at which CEO became CEO for the first time.
Financial expert CEO	CEO who has past experience in either banking or investment firms (two-digit SIC codes 60, 61, and 62), in a finance-related role (Accountant, CFO, Treasurer, or VP of Finance), or in a large auditing firm (current and former top-tier companies: Pricewaterhouse, Deloitte, Ernst & Young, KPMG, Arthur Andersen, Coopers, Peat Marwick, Touche Ross).
General Ability Index	First factor of applying principal components analysis to five proxies of general managerial ability: past number of positions, number of firms, number of industries, CEO experience, and conglomerate experience.
Ivy League alumnus	Graduated from an Ivy League college.
Law dummy	Dummy variable that takes the value of one if CEO has a law-related degree, and zero otherwise.
MBA dummy	Dummy variable that takes the value of one if CEO has an MBA degree, and zero otherwise.
Recession graduate	Graduated during NBER recession.
Science dummy	Dummy variable that takes the value of one if CEO has a science-related degree, and zero otherwise.
Sex dummy	A dummy equal to one if the CEO is male.
Panel B: Firm characteristics	
Variable	Definition
Acquisitions	Value of acquisitions (AQC).
Advertising	Advertising expenses (XAD).
Analyst dispersion	Standard deviation across different analysts of yearly earnings forecasts standardized by the assets of the firm (from I/B/E/S).
Assets	Book value of assets (AT).
Asset growth	Book value of assets (AT) in year t over book value of assets in year $t-1$, minus one.
Asset volatility	Standard deviation of stock return during the fiscal year times market value of equity (CSHO $\times$ PRCC_F) divided by market value of assets (AT + + CSHOXPRCC_F – CEQ).
Book leverage	Ratio of total debt (DLTT) to book value of assets (AT).
Capex	Ratio of capital expenditures (CAPX) to book value of assets (AT).
Cash	Ratio of cash and short-term investments (CHE) over book value of total assets (AT).
CFO turnover	A dummy variable that is equal to one if there is a CFO turnover.
Citations	Total number of citations received by the set of patents that a firm applied for in a given year from NBER patent database.
Citations (Q)	Adjusted total number of citations received by the set of patents that a firm applied for in a given year from NBER patent database. The adjustment uses the weighting index from Hall, Jaffe, and Trajtenberg (2001, 2005).
Debt issues	Dummy variable equal to one if the firm issues debt in a given year (from Dealscan).
Diversification dummy	Dummy variable that takes the value of one if a firm has more than one business segment and zero otherwise.
Dividend dummy	Dummy variable that takes the value of one if the firm pays dividends (DVC) and zero otherwise.
Firm age	Number of years between fiscal year (FYEAR) and CRSP listing year (LISTYEAR).
Firm size	Book value of assets (AT).
IOR	Institutional ownership using data from the Thomson CDA/Spectrum 13 F Holdings (see Aghion, Van Reenen, and Zingales, 2013).
Net working capital	Net current assets minus cash (ACT – CHE – LCT) divided by assets (AT) minus cash (CHE).
Patents	Number of patent applications from NBER patent database.
PPE	Ratio of net property, plant, and equipment (PPNT) to book value of assets (AT).
R&D	Ratio of research and development expenditures (XRD) to book value of assets (AT).
Rating dummy	Dummy variable that takes the value of one if a firm has a S&P domestic long-term issuer credit rating (SPLTCRM).
Repurchase dummy	Dummy variable equal to one if the firm buys back shares in a given year.
Retained earnings	Ratio of retained earnings (RE) and common equity (CEQ).
Return on assets (ROA)	Ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to book value of total assets (AT).
Sales	Sales (SALE).
Stock returns	Annual stock return $[(PRCC_F(t) / AJEX(t) + + DVPSX_F(t) / AJEX(t)) / (PRCC_F(t-1) / AJEX_F(t-1))]$.
Switch bank	Dummy variable that is equal to one if the leading loan provider of the first new loan after the CEO turnover did not offer the last loan before the turnover (or had not provided any loan in the last three years before the turnover).
Tobin's q	Ratio of market value of assets (AT + + CSHOXPRCC_F – CEQ) to book value of total assets (AT).
Total payout	Dividends plus share repurchases divided by net income.
Wage (average)	Total wage expense divided by the total number of employees in a given year.
Panel C: Market variables	
Variable	Definition
Default spread	Difference between the average yield on Moody's corporate bond ratings AAA and BAA.
Recession dummy	One if a given year is considered to be an NBER recession.

of cash and marketable securities to non-cash assets. The OLS regression includes interacted industry (two-digit SIC) and year fixed effects. The firm fixed effects regressions also include year fixed effects.

The control variables in these regressions are the firm characteristics typically used as determinants of cash holdings (see Opler, Pinkowitz, Stulz, and Williamson, 1999; Harford, Mansi, and Maxwell, 2008).¹¹ We find that cash is negatively related to the size of the firm, net working capital, and acquisitions, and positively related to investment opportunities proxied by Tobin's q , leverage, cash flow, the volatility of assets, and R&D expenses. Opler, Pinkowitz, Stulz, and Williamson (1999) and Bates, Kahle, and Stulz (2009) argue that this evidence is consistent with the idea that firms trade off the costs and benefits of cash holdings, but also with a precautionary motive for holding cash. Firms with better access to capital, such as large firms with more stable cash flows, tend to hold less cash.

In column 1 we run a cash holding regression in which we add the financial expert CEO dummy and other CEO characteristics such as age, age squared, sex, and tenure. In column 2 we run a similar specification with firm fixed effects relying only on within-firm variation; that is, our identification comes from cases in which a firm switches from a nonfinancial expert CEO to a financial expert CEO or vice versa. A financial expert CEO holds up to 12% less cash than a nonfinancial expert CEO. The coefficient of the financial expert CEO dummy is statistically significant at the 1% level in the OLS regression and at the 5% level in the firm fixed effects specifications, and is also economically meaningful.

Bates, Kahle, and Stulz (2009) argue the trend in corporate cash holdings over the last decades is mainly driven by the precautionary motive for holding cash. In general, our evidence is also consistent with firms holding cash for precautionary motives: CEOs with better access to financing—financial experts—seem to have fewer needs to hold large cash stocks.

Columns 3 and 4 of Table 3 show the results of regression tests of leverage on the financial expertise of the CEO. The dependent variable is the ratio of total debt to the book value of assets. We run an OLS regression with interacted industry (two-digit SIC) and year dummies and firm fixed effects regressions with year dummies. In both OLS and firm fixed effects regressions, t -statistics are adjusted for heteroskedasticity and within-firm correlation using clustered standard errors.

The set of controls includes firm size measured by the logarithm of assets' book value, Tobin's q as a measure of investment opportunities, the volatility of assets, research and development (R&D) as a proxy for intangibility, capital expenditures, a dummy variable for dividend-paying firms (see, for instance, Hovakimian, Opler, and Titman, 2001; or Lemmon, Roberts, and Zender, 2008), profitability measured by return on assets (ROA), and the tangibility of the assets measured by the ratio of property, plants, and equipment to total assets. We find leverage for the firms in our sample to be

positively associated with size and the tangibility of assets. We find negative and significant coefficients for asset volatility, capital expenditures, the dividend dummy, and ROA. These estimates are, for the most part, consistent with the results in Lemmon, Roberts, and Zender (2008), except that we do not find Tobin's q to be significant. These differences can possibly be explained by differences in the samples. The estimated coefficient for the financial expertise of the CEO is positive, which suggests that firms with this type of CEO tend to hold more debt. The impact of having a financial expert CEO on leverage is about 1.3 percentage points in the fixed effects regressions. This corresponds to 6% more leverage in relative terms for an average firm.

Columns 5 and 6 of Table 3 show the regression results of a linear probability model (LPM) of dividend payout on a financial expert CEO dummy and other CEO and firm-level controls. CEO-level controls include age, sex, tenure, education, talent, and general managerial ability. Firm-level controls include size, ROA, leverage, previous year dividend payout, asset growth, and cash (see DeAngelo, DeAngelo, and Stulz, 2006). We find a positive and significant coefficient of financial expertise in specification (5). The point estimate is 0.7%, corresponding to a 1.5% higher likelihood of paying out dividends on average.

When we include firm fixed effects, we do not observe any differences. This suggests that the differences between financial expert and nonfinancial expert CEOs in terms of dividend payout policy are mainly driven by omitted firm characteristics that are time-fixed. This might not be surprising, as dividend payout decisions are quite stable over time and something that may not require much financial sophistication.

Columns 7 and 8 of Table 3 show the results for share repurchases. The dependent variable in all regressions is a dummy variable equal to one if the firm buys back shares in a given year. Firm-level controls are firm size, lagged share repurchase dummy, leverage, profitability, cash, and asset growth. We find similar results across all specifications. Financial expert CEOs are more likely to repurchase shares than non-expert CEOs, controlling for the typical determinants of payout policy. Financial expert CEOs have a 2 to 4 percentage points higher propensity to engage in repurchases, which corresponds to a 7% higher likelihood, on average, in relative terms.

Overall, the results are consistent with the idea that financial expert CEOs follow different financial policies. The fact that financial expert CEOs hold less cash and more debt, and are more likely to buy back shares, is also consistent with a conclusion that they have easier access to external capital markets, a hypothesis we test below.

4. What do financial expert CEOs do?

In this section, we provide micro-level evidence on how financial expertise might operate.

4.1. Do financial expert CEOs follow financial theory more closely?

We know from research by Graham and Harvey (2001) that corporate finance practices differ across firms.

¹¹ In unreported tests, we run regressions of cash holdings on firm-level characteristics without including CEO financial expertise and other characteristics as controls. The coefficients obtained in these regressions are consistent with results in other studies.

Table 2

The sample consists of ExecuComp firms for which CEO profile data are available from BoardEx in the 1993–2007 period. All variables are winsorized at the 1st and 99th values. Variable definitions are as defined in Table 1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panel A: CEO financial expertise										
	Mean	p25	p50	p75	sd	N				
Financial expert dummy	0.411	0.000	0.000	1.000	0.492	17716				
Depository institutions	0.157	0.000	0.000	0.000	0.364	17710				
Nondep. institutions	0.048	0.000	0.000	0.000	0.214	17710				
Investment banks & securities brokerages	0.061	0.000	0.000	0.000	0.239	17710				
Auditing experience	0.074	0.000	0.000	0.000	0.261	17710				
CFO role	0.082	0.000	0.000	0.000	0.275	17710				
Financial role	0.030	0.000	0.000	0.000	0.170	17710				
Panel B: CEO characteristics										
	Mean	p25	p50	p75	sd	N	Financial expert CEO	Nonfinancial expert CEO	Diff.	S.E.
CEO age	55.434	50.000	55.000	60.000	7.773	17666	55.746	55.216	0.530***	(0.119)
Sex dummy	0.983	1.000	1.000	1.000	0.130	17716	0.981	0.984	−0.003	(0.002)
CEO tenure	7.277	2.000	5.000	10.000	7.522	16499	7.305	7.257	0.047	(0.119)
MBA dummy	0.371	0.000	0.000	1.000	0.483	17716	0.370	0.371	−0.001	(0.007)
Econ dummy	0.327	0.000	0.000	1.000	0.469	17716	0.308	0.340	−0.033***	(0.007)
Science dummy	0.353	0.000	0.000	1.000	0.478	17716	0.235	0.435	−0.201***	(0.007)
Law dummy	0.163	0.000	0.000	0.000	0.369	17716	0.090	0.214	−0.124***	(0.006)
Ivy League alumnus	0.172	0.000	0.000	0.000	0.378	17716	0.217	0.141	0.076***	(0.006)
Recession graduate	0.286	0.000	0.000	1.000	0.452	15485	0.311	0.265	0.045***	(0.007)
Generalist index	−0.009	−0.729	−0.182	0.544	0.975	15485	0.212	−0.194	0.406***	(0.015)
Fast track CEO	47.977	42.000	48.000	54.000	8.557	14227	48.159	47.824	0.334**	(0.144)
External hire dummy	0.403	0.000	0.000	1.000	0.490	15485	0.381	0.421	−0.039***	(0.008)
CEO-chair dummy	0.601	0.000	1.000	1.000	0.490	15485	0.654	0.557	0.097***	(0.008)
Total compensation	4155.889	1106.602	2232.096	4766.495	5539.798	15485	4584.638	3795.761	788.877***	(89.153)
Cash compensation	1173.025	556.732	878.828	1432.055	1002.469	15485	1281.427	1081.974	199.453***	(16.094)
Equity compensation	2398.365	120.705	818.158	2476.625	4473.618	13363	2618.260	2211.103	407.158***	(77.572)
Panel C: Firm characteristics										
	Mean	p25	p50	p75	sd	N	Financial expert CEO	Nonfinancial expert CEO	Diff.	S.E.
Cash	0.267	0.021	0.078	0.275	0.496	17716	0.200	0.314	−0.114***	(0.008)
Book leverage	0.216	0.049	0.200	0.328	0.182	17700	0.236	0.202	0.034***	(0.003)
Dividend dummy	0.481	0.000	0.000	1.000	0.500	17716	0.562	0.424	0.138***	(0.008)
Share repurchase dummy	0.578	0.000	1.000	1.000	0.494	17716	0.615	0.553	0.062***	(0.008)
Analyst dispersion	0.009	0.001	0.002	0.006	0.049	9951	0.005	0.012	−0.007***	(0.001)
Assets	4440.160	447.719	1101.703	3188.079	12104.040	17716	5994.446	3355.155	2639.291***	(183.76)
Tobin's q	2.137	1.263	1.665	2.425	1.443	17716	2.048	2.198	−0.15***	(0.022)
ROA	0.138	0.092	0.140	0.195	0.105	17715	0.144	0.134	0.01***	(0.002)
Asset volatility	0.276	0.146	0.220	0.346	0.191	17543	0.244	0.299	−0.055***	(0.003)
Firm age	21.091	8.000	15.000	30.000	18.371	17586	24.276	18.863	5.413***	(0.279)
Capex	0.063	0.027	0.047	0.081	0.055	17716	0.062	0.065	−0.003***	(0.001)
R&D	0.034	0.000	0.002	0.044	0.058	17716	0.026	0.040	−0.014***	(0.001)
High-tech dummy	0.205	0.000	0.000	0.000	0.404	17716	0.174	0.226	−0.052***	(0.006)
Asset growth	0.135	−0.012	0.071	0.191	0.312	15906	0.126	0.142	−0.016***	(0.005)
Acquisitions	0.035	0.000	0.000	0.032	0.079	17716	0.036	0.035	0.001	(0.001)
Diversification dummy	0.580	0.000	1.000	1.000	0.494	10559	0.600	0.564	0.036***	(0.01)
Net working capital	0.090	−0.016	0.091	0.213	0.198	17716	0.077	0.098	−0.022***	(0.003)
Retained earnings	0.360	0.195	0.570	0.867	1.589	17675	0.452	0.296	0.156***	(0.024)

Table 3

The dependent variable in regressions (1)–(2) is the log of cash and marketable securities. The dependent variable in regressions (3)–(4) is the ratio of total debt to assets. The dependent variable in regressions (5)–(6) is a dummy variable equal to one if the firm paid a dividend in the current year. The dependent variable in regressions (7)–(8) is a dummy variable equal to one if the firm repurchased shares in the current year. Variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Cash		Leverage		Dividends		Repurchases	
Financial expert CEO	−0.123*** (0.040)	−0.114** (0.051)	0.009* (0.005)	0.013** (0.006)	0.007** (0.003)	0.011 (0.009)	0.021*** (0.007)	0.036** (0.016)
Age	−0.030 (0.022)	−0.050* (0.026)	−0.000 (0.003)	−0.000 (0.003)	0.003 (0.002)	0.003 (0.004)	0.005 (0.005)	0.003 (0.009)
Age squared	0.000 (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	−0.000 (0.000)	−0.000 (0.000)	−0.000 (0.000)
Sex	−0.336* (0.199)	−0.073 (0.187)	0.018 (0.020)	−0.012 (0.017)	−0.004 (0.013)	−0.040 (0.038)	0.001 (0.027)	−0.026 (0.049)
Tenure	0.002 (0.003)	−0.008** (0.004)	−0.001* (0.000)	−0.000 (0.001)	0.000 (0.000)	0.000 (0.001)	−0.000 (0.001)	0.001 (0.002)
Log(assets)	−0.216*** (0.017)	−0.476*** (0.035)	0.013*** (0.002)	0.018*** (0.006)	0.010*** (0.001)	0.032*** (0.007)	0.027*** (0.003)	0.065*** (0.014)
Log q	0.626*** (0.035)	0.449*** (0.036)	−0.001 (0.007)	−0.008 (0.006)				
Asset volatility	1.239*** (0.107)	0.412*** (0.087)	−0.323*** (0.020)	−0.155*** (0.015)				
R&D	2.636*** (0.207)	0.601** (0.243)	0.031 (0.036)	−0.055 (0.050)				
Capex	−4.685*** (0.414)	−3.251*** (0.350)	−0.246*** (0.063)	−0.206*** (0.053)				
Dividend dummy	−0.143*** (0.048)	0.098* (0.053)	−0.048*** (0.006)	−0.025*** (0.008)				
ROA			−0.255*** (0.041)	−0.245*** (0.036)	0.157*** (0.018)	0.125*** (0.039)	0.486*** (0.039)	0.570*** (0.084)
PPE			0.153*** (0.025)	0.121*** (0.038)				
Leverage	0.079 (0.082)	0.305*** (0.079)			−0.055*** (0.011)	−0.085*** (0.025)	−0.219*** (0.024)	−0.201*** (0.050)
Cash flow	1.041*** (0.109)	0.468*** (0.084)						
Net working capital	−1.145*** (0.131)	−0.490*** (0.133)						
Acquisitions	−2.349*** (0.158)	−1.074*** (0.131)						
Dividend dummy($t-1$)					0.900*** (0.005)	0.605*** (0.019)		
Repurchase ($t-1$)							0.520*** (0.009)	0.251*** (0.013)
Asset growth					−0.001 (0.005)	−0.003 (0.007)	−0.082*** (0.012)	−0.098*** (0.017)
Cash					0.005 (0.003)	0.029*** (0.008)	0.050*** (0.009)	0.048** (0.022)
MBA		0.017 (0.054)		−0.001 (0.007)		−0.004 (0.010)		0.040** (0.018)
Econ		−0.051 (0.054)		0.004 (0.007)		−0.010 (0.010)		−0.017 (0.019)
Science		−0.093 (0.059)		0.004 (0.008)		0.000 (0.010)		−0.025 (0.021)
Law		−0.011 (0.083)		0.009 (0.010)		−0.023 (0.018)		−0.008 (0.037)
Ivy League alumnus		−0.015 (0.063)		0.008 (0.008)		−0.010 (0.010)		−0.018 (0.020)
Generalist index		0.034 (0.028)		−0.001 (0.003)		−0.004 (0.004)		0.008 (0.009)
Recession graduate		0.006 (0.050)		−0.001 (0.006)		−0.010 (0.009)		0.003 (0.018)
Fast track CEO		−0.003 (0.003)		0.000 (0.000)		0.000 (0.001)		0.001 (0.001)
Obs	16191	13594	16275	13667	14138	11986	14138	11986
Industry dummies	Yes	No	Yes	No	Yes	No	Yes	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
R-squared	0.552	0.789	0.319	0.718	0.873	0.890	0.127	0.437

Their surveys identify aspects of corporate practice that are consistent with finance theory and others that are hard to reconcile with it. With respect to capital budgeting, most companies follow academic theory and use discounted cash flow (DCF) and net present value (NPV) techniques to evaluate new projects, but many use companywide, rather than project-specific, discount rates to evaluate opportunities. Additionally, when it comes to making capital structure decisions, corporations appear to pay less attention to finance theory and rely instead on practical, informal rules of thumb.¹²

Financial expert CEOs might have been more exposed to finance theory over their career than non-experts, which could increase their ability or sophistication to apply them in practice or made them aware of such theories in the first place.

Krüger, Landier, and Thesmar (2011, p. 1) “document investment distortions induced by the use of a single discount rate within firms. According to textbook capital budgeting, firms should value any project using a discount rate determined by the risk characteristics of the project. If they use a unique company-wide discount rate, they overinvest (resp. underinvest) in divisions with a market beta higher (resp. lower) than the firm's core industry beta. We directly test this consequence of the “WACC fallacy” and establish a robust and significant positive relationship between division-level investment and the spread between the division's market beta and the firm's core industry beta.”

We first replicate their main result using our sample. We then analyze whether financial expert CEOs are less likely to fall into the trap of the WACC fallacy. Table 4 shows the results. Column 1 replicates the key result of Table 5 of Krüger, Landier, and Thesmar (2011): Non-core division raw investment depends positively on the spread between the non-core and the core division's industry betas. We use the average weighted average cost of capital (industry level) that is provided in Table A.I in Krüger, Landier, and Thesmar (2011).

In columns 2 and 3 we split the sample and report results for financial expert CEOs and non-financial expert CEOs separately. The coefficient on the delta spread is more than twice in magnitude (0.028 vs. 0.012) and only statistically different from zero for nonfinancial expert CEOs. This result suggests that the evidence on WACC fallacy is mainly driven by non-financial expert CEOs.

To understand the economic magnitude of this effect, we follow the interpretation and assumptions in Krüger, Landier, and Thesmar (2011). Assume the difference between the beta of the non-core division and the beta of the core division to be 0.31, which is about one sample standard deviation. This means the gap in discount rates between the division and its core is approximately 2.2% (assuming a 7% equity risk premium). Given our estimates for non-experts, we would expect the non-core division's investment rate to be 0.9 (0.028×0.31) percentage points higher (using the estimated coefficient in column 2). This

is equivalent to about 16% of the average investment rate of 0.057 in our sample of non-core divisions.¹³

In columns 4 through 7 we add further firm controls as well as CEO controls. The results are robust to the inclusion of these controls.

In short, we replicate the results in Krüger, Landier, and Thesmar (2011) on the WACC fallacy, but we show that their result is mostly driven by firms run by CEOs without finance experience. This evidence supports the view that financial expert CEOs are more aware of financial theory and are more financially sophisticated.

4.2. Do financial expert CEOs communicate better with outside investors?

One potential mechanism through which CEOs get better access to credit markets is improved communication between firm management and outside investors, who are mainly financial experts themselves (e.g., loan officers or financial analysts). Having a similar professional background might mean the CEOs speak the same language and can communicate more effectively when funding is needed.

There is evidence that investors are more likely to hold stocks they are familiar with in terms of their professional background or that are closer in terms of language and culture (see Massa and Simonov, 2006). Grinblatt and Keloharju (2001) find investors in Finland prefer to hold and trade firms whose CEO is of similar cultural origin. These findings could be consistent with behavioral biases or with better (more) information, but also with greater ability to evaluate and interpret information by referring to the same technical language and to the same underlying theories. There is also evidence of the importance of soft information in financial intermediation; examples are Stein (2002), Berger, Miller, Petersen, Rajan, and Stein (2005), and Cerqueiro, Degryse, and Ongena (2011). Analyzing a large sample of German customer loans, Puri, Rocholl, and Steffen (2011) show that loan officer discretion is widespread.

To explore the channel of improved communication, we analyze firms' communication with financial analysts. We obtain data on earnings forecasts by financial analysts from I/B/E/S. We define the dispersion of the analyst forecasts as the standard deviation across different analysts of yearly earnings forecasts standardized by the assets of the firm. This measure should capture the level of disagreement among different analysts on the earnings perspective of a company, i.e., how effective the communication of the management was. Table 5, Panel A shows that the average dispersion of analyst earnings forecasts is 0.012 if the CEO is not a financial expert, which is more than double the forecast dispersion of firms with financial expert CEOs (0.005). The difference is also statistically significant (t -stat of 7.04). We interpret this result as first evidence that financial expert CEOs are better communicators of information that is relevant for financial markets.

In columns 1 and 2 of Panel B of Table 5, we test for differences in analyst dispersion forecasts in a multivariate setting. The dependent variable is, as before, the

¹² See Graham and Harvey (2001, p. 189) and Graham and Harvey (2002, pp. 8–9).

¹³ In their sample, they estimate 10% compared to 16% in our sample.

Table 4

This table shows the effect of financial experience on potential overinvestment (underinvestment) in high (low) beta division because of applying a companywide discount rate instead of a division-specific one. We follow Krüger, Landier, and Thesmar (2011), Table 5. "Using business segment data from Compustat, we construct industry-level divisions by aggregating segment. A division is defined as the bundle of arm's segments operating in the same FF48 industry. The regressions are run on divisions which do not have the highest sales in the conglomerate (non-core divisions). The dependent variable Capex/Assets is division-level capital expenditures in period $t+1$ scaled by division assets in period t . Delta beta is the difference between the industry-level asset beta of the core-division (i.e., the division with the highest sales) and the industry-level asset beta of the non-core division. Division q is the division's industry-level Tobin's q . Core q is the industry-level Tobin's q of the division with the highest sales in the conglomerate. Both are calculated for a sample of standalone firms from the same industry. Firm cash flow is the firm's cash flow scaled by total assets. Division size is the logarithm of division sales and Firm size is the log of total assets. Firm age is the logarithm of the current year plus one minus the year in which the firm first appeared in the Compustat North America database." CEO and firm variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Capex(non-core)						
Delta beta	0.020*** (0.006)	0.028*** (0.008)	0.012 (0.008)	0.026*** (0.007)	0.008 (0.008)	0.020*** (0.007)	0.009 (0.009)
Segment q				0.000 (0.000)	−0.000* (0.000)	0.000 (0.000)	−0.000 (0.000)
Core q				−0.000** (0.000)	−0.000 (0.000)	−0.000** (0.000)	0.000 (0.000)
Segment size				−0.003 (0.002)	−0.008*** (0.002)	−0.002 (0.003)	−0.009*** (0.003)
Cash flow				0.025 (0.018)	0.042* (0.023)	0.008 (0.021)	0.035* (0.020)
Firm size				0.000 (0.004)	0.005* (0.003)	−0.002 (0.004)	0.006** (0.003)
Firm age				0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Asset growth						0.009 (0.010)	−0.003 (0.006)
Retained earnings						−0.001 (0.003)	−0.000 (0.001)
Age						−0.001 (0.002)	0.002 (0.003)
Age squared						0.000 (0.000)	−0.000 (0.000)
Gender						−0.026 (0.024)	0.047* (0.027)
Tenure						0.000 (0.000)	0.000 (0.000)
Econ						0.006 (0.005)	0.011** (0.005)
Science						0.005 (0.005)	0.008 (0.006)
Law						−0.017** (0.007)	0.004 (0.006)
MBA						0.001 (0.005)	−0.007 (0.005)
Ivy League alumnus						0.005 (0.005)	0.002 (0.005)
Generalist index						−0.000 (0.002)	0.001 (0.002)
Recession graduate						0.005 (0.005)	−0.005 (0.004)
Fast track CEO						−0.000 (0.000)	0.000 (0.000)
Sample	All	Nonfinancial experts	Financial experts	Nonfinancial experts	Financial experts	Nonfinancial experts	Financial experts
Obs	4976	2564	2412	2549	2401	1717	1972
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-squared	0.102	0.121	0.108	0.130	0.123	0.159	0.122

dispersion of analyst forecasts and the main variable of interest is a dummy variable that is equal to one if the CEO is a financial expert. We find a negative and significant effect of −0.002 (corresponding to a reduction of about

25% on average) when controlling for firm characteristics, basic CEO characteristics, and industry fixed effects. We do not observe any differences, however, when including firm fixed effects in the specification.

Table 5

This table shows the effect of financial experience on the dispersion of analyst forecasts. The dependent variable in all regressions is standard deviation of yearly earnings forecasts divided by non-cash assets. CEO and firm variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panel A: Univariate analysis			
Financial expert CEO	Nonfinancial Expert CEO	Diff.	s.e.
0.005	Analyst forecast dispersion 0.012	–0.007***	(0.001)
Panel B: Multivariate analysis			
	(1)	Analyst forecast dispersion	(2)
Financial expert CEO	–0.002** (0.001)	0.000 (0.000)	
Log(assets)	–0.007*** (0.001)	–0.006*** (0.002)	
Firm age	0.000** (0.000)	0.000 (0.000)	
Tobin's q	0.005 (0.004)	–0.003 (0.002)	
Asset growth	–0.008*** (0.002)	–0.002 (0.002)	
Retained earnings	–0.000 (0.001)	–0.001* (0.000)	
Asset volatility	0.015 (0.020)	0.007 (0.004)	
R&D	0.026 (0.022)	0.043 (0.033)	
Age	0.000 (0.001)	–0.001 (0.001)	
Age squared	–0.000 (0.000)	0.000 (0.000)	
Sex	0.001 (0.004)	0.001 (0.001)	
Tenure	–0.000 (0.000)	0.000 (0.000)	
MBA		0.000 (0.001)	
Econ		–0.001 (0.001)	
Science		–0.001 (0.001)	
Law		–0.001 (0.001)	
Ivy League alumnus		–0.001 (0.001)	
Generalist index		0.000 (0.000)	
Recession graduate		–0.001 (0.001)	
Fast track CEO		0.000 (0.000)	
Obs	8128		7462
Industry dummies	Yes		No
Year dummies	Yes		Yes
Firm fixed effects	No		Yes
Adjusted R -squared	0.087		0.529

From the results in this section we conclude that there is some evidence that financial expert CEOs are better at communicating firm prospects with other entities in financial markets, as reflected by lower dispersion in analysts' forecasts for their firms.

4.3. Does financial expertise decrease investment-to-cash flow sensitivity?

We test if firms with financial expert CEOs benefit from lower sensitivity of investment to cash flows. This would

be a consequence of better access to financing by this type of CEO. Table 6 shows the results. We run OLS and firm fixed effects regressions of investment, measured by the ratio of capital expenditures-to-assets, cash flow, the financial expert dummy, the interaction of cash flow with the financial expert dummy, and a set of additional controls following Guner, Malmendier, and Tate (2008). We include rating dummies, rating dummies interacted with cash flows, year dummies, and year dummies interacted with cash flows, as well as industry dummies and the interaction between industries and cash flows.

Specifications (1) and (2) follow the same controls as in [Guner, Malmendier, and Tate \(2008\)](#). Specifications (3) and (4) include further CEO characteristics as controls.

The main coefficient of interest is the interaction term between the financial expertise of the CEO and cash flow. We find this coefficient to be negative and significant across all specifications. This result indicates that investments of firms with financial expert CEOs are less sensitive to internally generated cash flows. This result is in line with [Guner, Malmendier, and Tate \(2008\)](#) who find that nonfinancial firms with financial expert independent board members need to rely less on internally generated funds to invest. The result is also consistent with [Malmendier and Tate \(2005\)](#) who show that CEOs with technical education have more investment-cash flow sensitivity than CEOs with general education while CEOs with financial education have less. [Guner, Malmendier, and Tate \(2008\)](#), however, also find that the firms benefiting from this effect are mainly firms with poor investment opportunities, which leads to inefficient overinvestment. Their reasoning is that bankers on the boards of these firms have an incentive to benefit their own financial institutions, and thus generate business by providing funds to the firms. Such a conflict should not be the case for a CEO who is a financial expert, with no apparent reason to benefit any financial institution.

We do not find any evidence consistent with a conclusion that financial expert CEOs over-invest. The coefficient of financial expertise of the CEO in [Table 6](#) is not statistically significant in specifications (1) and (3) and very small economically in specifications (2) and (4). This suggests that financial expert CEOs do not invest significantly more than nonfinancial experts, which is not consistent with a hypothesis of overinvestment. Note that we control for industry average investment in all regressions by adding industry fixed effects and for investment opportunities by adding Tobin's q .¹⁴

The results so far add to our previous findings that firms can benefit from appointing a financial expert CEO because they are better able to raise external finance. We find not only that this expertise can make investment less sensitive to internal resources of the firm, but also that it is not associated with inefficient investment. These results are in stark contrast to the findings of [Guner, Malmendier, and Tate \(2008\)](#), and suggest that appointing a financial expert CEO is probably a better way to benefit from financial expertise than to risk the costs of having a banker on the board.¹⁵

4.4. How do financial expert CEOs react to shocks to overall credit conditions?

To analyze financial policies from a dynamic perspective, we test whether financial expert CEOs react differently to changes in the market environment and whether

they manage financial policies more actively. We exploit exogenous variation in overall credit conditions for firms, measured by the default spread (the difference between the average yield on Moody's corporate bond ratings Aaa and Baa).¹⁶ [Table 7](#) shows the results of regressing cash holdings on the default spread and interaction of the default spread with the financial expert dummy. Firm-level control variables in these regressions are the same as the ones in [Table 3](#).

The main independent variable of interest is the interaction term of the default spread and the financial expert CEO dummy. The coefficient of this interaction is positive and statistically significant at the 5% level in column 1 and the 10% level in column 2. This result is consistent with the idea that financial expert CEOs can increase cash holdings when external markets discourage the raising of cash.

In untabulated tests we run regressions of cash holdings on the default spread, a recession dummy, and other firm-level determinants without including the financial expert CEO dummy. The recession dummy has a negative effect which turns insignificant once we control for firm fixed effects, and the default spread has a positive effect suggesting that firms increase their cash holdings at times when external financing is more costly.¹⁷ This result is consistent with [Almeida, Campello, and Weisbach \(2004\)](#) and [Bates, Kahle, and Stulz \(2009\)](#) who find evidence of a precautionary motive for cash holdings.

At this point we cannot reject the hypothesis that the increase in cash holdings when the default spread is high comes from internally generated funds that are retained in the firm. In the next tests we go one step further to see whether financial expert CEOs are able to issue debt when the default spread is high.

Specifications (3) to (4) in [Table 7](#) test whether financial expert CEOs are better able to raise debt when it is more difficult to get access to credit. A limited probability model includes the same firm-level controls as in [Table 3](#). The dependent variable in all regressions is a dummy variable equal to one if the firm issues debt in a given year. The debt issues are obtained from Dealscan. The main variable of interest is interaction of the default spread with the financial expert dummy. The coefficient of this interacted variable is positive and significant at the 10% level in the main specifications but turns insignificant when controlling for firm fixed effects.

The fact that financial expert CEOs hold more cash and are not so constrained in accessing debt markets in periods of high default spreads suggests benefits associated with financial expertise. Because we find very weak results with respect to CEO education, this suggests that the operant mechanism is related to CEO professional experience. It might be that financial expert CEOs have better access to a financial network and to financing providers, a hypothesis that we test below. Or, it could be that through their career path, financial expert CEOs have

¹⁴ In [Guner, Malmendier, and Tate \(2008\)](#), however, the coefficient for financial expertise of the board, particularly of commercial bankers, is positive and significant across specifications.

¹⁵ We also test whether financial expertise of the CEO is linked to higher financing costs by looking at interest payments. We do not find this to be the case.

¹⁶ If shocks to credit conditions, measured by the default spread, are unexpected by firms, the matching between CEOs and firms could be considered to be exogenous to this shock.

¹⁷ See [Korajczyk and Levy \(2003\)](#) for the effects of macroeconomic conditions on firm leverage.

Table 6

The dependent variable in regressions (1) to (4) is the ratio of capital expenditures to assets. Variable definitions are included in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)
			Capex	
Fin. exp. CEO × Cash flow	−0.049*** (0.015)	−0.044*** (0.015)	−0.056*** (0.015)	−0.050*** (0.015)
Financial expert CEO	0.001 (0.002)	0.005** (0.002)	0.002 (0.002)	0.006** (0.002)
Tobin's q ($t-1$)	0.001 (0.000)	0.003*** (0.001)	0.000 (0.000)	0.002*** (0.000)
Log assets ($t-1$)	−0.003*** (0.001)	−0.011*** (0.002)	−0.003*** (0.001)	−0.011*** (0.002)
Log assets × Cash flow ($t-1$)	0.017*** (0.001)	0.012*** (0.001)	0.016*** (0.001)	0.012*** (0.001)
Board size	−0.009* (0.005)	−0.006 (0.005)	−0.004 (0.005)	−0.002 (0.005)
Board size × Cash flow	0.064 (0.040)	0.015 (0.034)	0.033 (0.037)	0.008 (0.035)
Board independence	−0.003 (0.006)	−0.007 (0.007)	0.000 (0.006)	−0.010 (0.007)
Board independence × Cash flow	0.060 (0.049)	0.069 (0.052)	0.038 (0.054)	0.109* (0.056)
Age			−0.002** (0.001)	−0.001 (0.001)
Age × Cash flow			0.005 (0.011)	0.001 (0.010)
Age squared			0.000 (0.000)	0.000 (0.000)
Age squared × Cash flow			−0.000 (0.000)	−0.000 (0.000)
Sex			0.008 (0.005)	−0.005 (0.005)
Sex × Cash flow			−0.026 (0.044)	−0.005 (0.026)
Tenure			0.000 (0.000)	0.000 (0.000)
Tenure × Cash flow			0.001 (0.001)	0.001 (0.001)
MBA			−0.002 (0.002)	0.001 (0.002)
MBA × Cash flow			0.017 (0.015)	−0.004 (0.015)
Econ			0.001 (0.002)	−0.001 (0.003)
Econ × Cash flow			0.007 (0.019)	0.011 (0.020)
Science			0.001 (0.002)	−0.000 (0.003)
Science × Cash flow			0.008 (0.019)	−0.003 (0.021)
Law			−0.008** (0.004)	−0.007* (0.004)
Law × Cash flow			0.028 (0.032)	0.011 (0.027)
Ivy League alumnus			0.003 (0.002)	0.002 (0.003)
Ivy League alumnus × Cash flow			−0.006 (0.016)	−0.019 (0.019)
Generalist index			−0.000 (0.001)	−0.000 (0.001)
Generalist × Cash flow			−0.011 (0.009)	0.002 (0.010)
Recession graduate			−0.000 (0.002)	−0.000 (0.003)
Recession graduate × Cash flow			0.014 (0.017)	−0.012 (0.017)
Obs	10538	10538	9298	9298
Rating dummies	Yes	Yes	Yes	Yes
Rating dummies × Cash flow	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes

Table 6 (continued)

	(1)	(2)	(3)	(4)
Year dummies × Cash flow	Yes	Yes	Yes	Yes
Industry dummies	Yes	No	Yes	No
Industry dummies × Cash flow	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	Yes
Adjusted R-squared	0.515	0.722	0.733	0.781

acquired the necessary skills to facilitate access to capital markets, perhaps through better ability to communicate with the markets by reducing information asymmetries.

Overall, the results in this section suggest that financial expert CEOs follow more dynamic financial policies, and are able to raise financing in times of worse credit conditions..

4.5. Are financial expert CEOs more responsive to tax cuts?

In this section, we further test whether financial expert CEOs respond to changes in the business environment in a way that indicates financial sophistication. We use the dividends and capital gains tax cuts in the Job and Growth Tax Relief Reconciliation Act of May 2003, also known as the Bush Tax Cuts. After these tax cuts, the after-tax value of dividends and capital gains to individual investors increased significantly, as the dividend tax dropped from 38.6% to 15% and the long-term capital gains from 20% to 15%. Firms efficiently reacted to these changes by increasing payout to shareholders, especially in the form of dividends (see, for instance, [Chetty and Saez, 2005](#); [Brown, Liang, and Weisbenner, 2007](#)).¹⁸ We expect financial expert CEOs to be more responsive to these changes, as they should better understand their implications. This test also helps to sort out the selection concerns, as firms are unlikely to change their CEO type to a financial expert with the sole purpose of responding to the tax changes.

We test whether financial experts pay out more to shareholders after the tax change. The dependent variable in our regressions is total payout (dividends and share repurchases) standardized by net income. We focus on total payout because this is a cleaner test. While it is clear that firms should increase payout to shareholders irrespective of its form (both taxes on dividends and capital gains decreased), it is not clear whether firms should increase payout via dividends or share repurchases. On the one hand, the tax disadvantage of dividends with respect to share repurchases was substantially reduced, and therefore, some firms might choose to increase payout through dividends. On the other hand, the tax cut was meant to be temporary and firms avoid cutting dividends as this tends to be perceived by the market as bad news. For this reason, some firms might decide to increase share repurchases instead of increasing dividends to prevent having to cut them later on.

¹⁸ In untabulated results we replicate in our sample the existing evidence that firms, in general, increased the payout, both in the form of dividends and share repurchases after 2003.

Results are shown in columns 5 and 6 of [Table 7](#). The coefficient of interest is the interaction term of the *Post-2003 dummy* with the *Financial Expert dummy*, which is positive and significant suggesting that financial experts are more responsive to the tax cut and pay out more to shareholders. The effect is economically relevant. The coefficient in specification (5) is 0.105, which represents an increase of 17% with respect to a mean payout of 0.59.

Overall, these results are consistent with previous evidence that financial expert CEOs are more dynamic in reacting to changes in the business environment in a way that suggests them being more financially sophisticated.

4.6. Do financial expert CEOs actively choose CFOs?

There is evidence that capital structure decisions are made mainly by CEOs, with execution delegated to CFOs. In a survey, [Graham, Harvey, and Puri \(2011\)](#) find that about 15% of responding CEOs claim that decisions on capital structure are made solely by themselves and only 2% of CEOs claim they are not involved in the decision making at all. Only 3% of responding CFOs say they are solely responsible for setting capital structure, while about 18% do not seem to have any role at all. As CFOs are still likely to play some part in setting and managing financial policies, we are interested in whether financial expert CEOs manage CFOs (and their replacement) differently.

Financial expert CEOs might, by their nature, be more interested in getting involved in financial policies, and potentially in changing them. The strategic management literature describes the replacement of top executives as a mechanism for an organization to overcome inertia ([Tushman and Romanelli, 1985](#)) or loyalty to a former CEO and to adapt strategically to changing contexts ([Helmich, 1977](#); [Pfeffer and Salancik, 1978](#); [Vancil, 1987](#)).¹⁹ Because executives are constrained by their experiences ([Hambrick and Mason, 1984](#)), new CEOs may want to replace executives whose knowledge and skills have become obsolete to facilitate desired strategic changes ([Keck and Tushman, 1993](#); [Wiersema and Bantel, 1993](#)).²⁰

¹⁹ [Shen and Cannella \(2002\)](#) argue that post-succession senior executive turnover reflects a new CEO's efforts to clear executive deadwood and to facilitate strategic reorientation ([Friedman and Saul, 1991](#); [Helmich and Brown, 1972](#); [Kesner and Dalton, 1994](#); [Wiersema, 1995](#)).

²⁰ An alternative explanation is that incumbent CFOs are more likely to resign on their own if a financial expert CEO should be hired. A financial expert CEO might be anticipated as possibly limiting a CFO's area of responsibility, which would diminish job satisfaction.

Table 7

The dependent variable in columns 1 and 2 is the log of cash to assets. The dependent variable in columns 3 and 4 is a dummy variable equal to one if the firm issues debt in a given year. The dependent variable in columns 5 and 6 is the sum of dividends and share repurchases normalized by net income. All regressions include industry and year dummies. Variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Cash holdings		Debt issues		Total payout	
Financial expert CEO × Default spread	0.267** (0.128)	0.218* (0.112)	0.074* (0.041)	0.051 (0.044)		
Financial expert CEO × Post-2003					0.105** (0.046)	0.092* (0.054)
Financial expert CEO	−0.345*** (0.114)	−0.295*** (0.109)	−0.051 (0.035)	−0.002 (0.042)	0.037 (0.028)	0.079 (0.049)
Age	−0.029 (0.022)	−0.051* (0.026)	0.006 (0.007)	−0.009 (0.010)	0.010 (0.013)	−0.021 (0.026)
Age squared	0.000 (0.000)	0.000** (0.000)	−0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)
Sex	−0.336* (0.199)	−0.074 (0.187)	0.045 (0.039)	−0.026 (0.061)	−0.003 (0.102)	−0.048 (0.191)
Tenure	0.002 (0.003)	−0.008** (0.004)	−0.000 (0.001)	−0.000 (0.002)	−0.002 (0.002)	0.009* (0.005)
Log(assets)	−0.215*** (0.017)	−0.474*** (0.035)	0.067*** (0.004)	0.102*** (0.014)	0.042*** (0.010)	0.077*** (0.036)
Log q	0.625*** (0.035)	0.450*** (0.036)				
Leverage	0.078 (0.082)	0.305*** (0.078)			−0.115 (0.081)	0.433*** (0.158)
Cash flow	1.040*** (0.109)	0.466*** (0.084)				
Asset volatility	1.240*** (0.107)	0.411*** (0.087)	−0.218*** (0.034)	−0.032 (0.038)		
Net working capital	−1.148*** (0.131)	−0.494*** (0.133)				
R&D	2.638*** (0.207)	0.610** (0.244)				
Capex	−4.686*** (0.414)	−3.265*** (0.350)	0.940*** (0.108)	0.840*** (0.147)		
Acquisitions	−2.348*** (0.158)	−1.074*** (0.131)				
Dividend dummy	−0.141*** (0.048)	0.100* (0.053)				
MBA		0.016 (0.054)		−0.009 (0.018)		0.048 (0.051)
Econ		−0.051 (0.054)		0.009 (0.018)		−0.088* (0.047)
Science		−0.093 (0.059)		0.015 (0.021)		−0.101 (0.062)
Law		−0.010 (0.083)		0.031 (0.029)		−0.027 (0.095)
Ivy League alumnus		−0.014 (0.063)		−0.007 (0.022)		−0.011 (0.059)
Generalist index		0.035 (0.028)		−0.005 (0.009)		0.014 (0.025)
Recession graduate		0.007 (0.050)		0.009 (0.018)		0.036 (0.049)
Fast track CEO		−0.003 (0.003)		0.001 (0.001)		0.006* (0.004)
Asset growth					−0.344*** (0.041)	−0.340*** (0.052)
Obs	16191	13594	16295	13682	13184	11169
Industry dummies	Yes	No	Yes	No	Yes	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	Yes	No	Yes
Adjusted R -squared	0.552	0.789	0.246	0.436	0.069	0.037

We restrict the analysis to the replacement of CFOs after CEO turnovers. We define a dummy variable that is equal to one if the CFO is also replaced within the first year

of the new CEO's tenure. In a multivariate analysis, we examine whether it is more likely the CFO will be replaced if the incoming CEO has financial expertise. In column 1 of

Table 8, we estimate a linear probability model (LPM) including year and industry dummies and company- and CEO-level variables as controls. We find a strong positive effect of the financial expertise of the CEO on CFO turnover. The likelihood of replacing the CFO is about 7% higher if the incoming CEO is a financial expert.

Financial expert CEOs might not be directly able to manage a firm's financial policies more efficiently, but they could be better at identifying weak CFOs and replacing them with more talented ones. Hence, financial expert CEOs might replace CFOs differently from nonfinancial experts. In untabulated analyses, we test this hypothesis relying on some observable measures of talent of CFOs. We analyze whether—conditional on a CFO being replaced—newly hired CFOs by financial expert CEOs are more talented than CFOs hired by nonfinancial experts. The proxies for CFO talent include previous CFO experience, CFO tenure, age of first appointment to CFO, investment banking experience, auditing experience, or consulting experience (Aier, Comprix, Gunlock, Lee, 2005; Ge, Matsumoto, and Zhang, 2011). We do not find any statistically significant difference in any dimension analyzed between CFOs who were hired by financial expert CEOs and non-experts. The sample is small, therefore not finding any differences might reflect a power problem.

Overall, our findings are consistent with the idea that financial expert CEOs become more involved than others in the financial policies of a firm. As it is likely to be easier to implement changes in a firm by changing management (avoiding entrenched habits), financial expert CEOs are more likely to replace the CFO. Consistent with Graham, Harvey, and Puri (2011), who show that CFOs mostly execute tasks delegated by CEOs, the identity or characteristics of the CFOs do not seem to matter much.

4.7. Do Financial Expert CEOs Rely on Their Professional Networks?

A potentially important part of CEO human capital is his network. Financial expert CEOs would want their firms to benefit directly from their links to the financial sector. An example is preferential financing from a former employer. We test this hypothesis by examining whether former employers of CEOs are providing loans to their companies. We construct the loan sample as follows. First, we identify all loan facilities obtained by the companies in our sample, which leads to 11,139 observations. Following Bharath, Dahiya, Saunders, and Srinivasan (2011), we then identify the banks that had lead positions, by restricting the involved banks to institutions that were “agent,” “administrative agent,” “arranger,” or “lead bank.” If several banks have lead-arranger roles, we include them all. We obtain 14,476 different loan facility-lender combinations including 674 different financial institutions (measured by Dealscan's companyid). Many financial institutions, however, are part of the same lending company. The different companyids occur because of different spelling or inclusion of branches of the same company (e.g., JPMorgan Chase Bank, JP Morgan Chase Bank NA). Aggregation gives us 369 unique leading lenders in the

sample. We analyze the CVs of the CEOs (from the BoardEx database) to identify people who worked for these lending institutions. We include all positions (executive and non-executive) and subsidiaries in similar fashion as when aggregating the leading banks. We find that 392 CEOs (10% of our sample) have worked in 110 (of 369) leading lending institutions. As we are interested in whether a particular loan facility was by a former employer of the CEO, we analyze every facility for potential links between the CEO and the lending institution. We find 242 loan facilities (or 2.17%) with a link.

Work experience in the banking sector might have allowed CEOs to develop a wide network in the financial industry beyond their former employers. To address this indirect channel, we test whether recently hired financial expert CEOs are more likely to switch lenders (i.e., change to a bank they are connected with). Observing a higher likelihood of switching loan providers by financial expert CEOs would be consistent with a network effect; it is also consistent with the idea that financial expert CEOs have superior knowledge about banking that allows them to select banks more efficiently. On the contrary, if we observe that financial expert CEOs are not more likely to change loan providers, it is less plausible that the CEO's network is driving the results. To test whether financial expert CEOs are more likely to switch loan providers, we restrict the sample to loans to companies that recently changed CEOs. We analyze the first loan obtained under the management of the new CEO, and check whether there had been an earlier relationship between the company and the bank providing the new loan. We look at the last loan before the CEO turnover and also at all deals in the last three years before the turnover to determine whether there has been a relationship. The results are reported in columns 2 and 3 of **Table 8**. The dependent variable is a dummy variable that is equal to one if the leading loan provider of the first new loan after the CEO turnover did not offer the last loan before the turnover (column 2) or had not provided any loan in the last three years before the turnover (column 3). We do not find financial expert CEOs are more likely to move to a new bank after their appointment. While the point estimates are very low in absolute terms and not statistically different from zero, the test is not powerful. With so few observations, not finding any difference might be a pure statistical artifact.

In short, the results in this section do not offer support for the view that financial experts have better access to financial markets because they rely on their personal networks. We do not find evidence that financial experts rely on their past employers to raise financing. We also do not find any evidence that they are more likely to change banks when raising debt.

5. Identification and alternative hypotheses

In our primary analysis, we follow the majority of the literature, which relies (at most) on within-firm variation in estimating CEO effects. The findings of the previous sections are consistent with a causal effect of financial expertise on firm policies.

Table 8

Column 1 shows the effect of financial experience on CFO turnover. The dependent variable in this regression is a dummy variable which is equal to one if the CFO is replaced in the first year of tenure of a new CEO. CEO and firm controls are CEO age, CEO age squared, sex, tenure, log assets, log q , book leverage, cash flow, asset volatility, net working capital, R&D, capex, acquisition dummy, and dividend dummy. Columns 2 and 3 show the effect of financial experience on choice of lending institutions. The dependent variable in these regressions is a dummy variable that is equal to one if the CEO is obtaining the first loan facility after appointment from a bank the company was not doing business with before. In column 2 only the last loan facility before the appointment is considered. In column 3 we consider all loan facilities in the last three years before the CEO was appointed. Variable definitions are as defined in the Appendix. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)
	CFO turnover	Switch banks	
	CFO is replaced within the first year after CEO turnover	Different from last bank before CEO turnover	Different from 3 years before CEO turnover
Financial expert CEO	0.070** (0.034)	0.015 (0.049)	–0.051 (0.049)
Age	–0.007 (0.028)	–0.032 (0.036)	–0.045 (0.032)
Age squared	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Sex	–0.165* (0.088)	–0.329* (0.190)	–0.210 (0.157)
Tenure	0.009 (0.025)	0.090*** (0.026)	0.047* (0.028)
Log(assets)	0.016 (0.013)	–0.035* (0.018)	–0.048** (0.019)
Log q	–0.042 (0.035)	–0.100* (0.060)	–0.091 (0.058)
Leverage	0.068 (0.116)	–0.054 (0.175)	–0.244 (0.175)
Cash flow	–0.195** (0.096)	0.123 (0.135)	0.149 (0.128)
Asset volatility	0.040 (0.124)	0.198 (0.216)	0.122 (0.210)
Net working capital	0.099 (0.103)	–0.260 (0.178)	–0.372** (0.173)
R&D	–0.153 (0.188)	0.488 (0.541)	0.726 (0.597)
Capex	–0.189 (0.409)	–1.628** (0.693)	–0.685 (0.743)
Acquisitions	–0.160 (0.226)	–0.141 (0.377)	–1.017*** (0.350)
Dividend dummy	–0.022 (0.037)	0.008 (0.055)	–0.032 (0.055)
Log(cash)	0.022 (0.013)	0.003 (0.020)	–0.007 (0.020)
MBA	0.027 (0.033)	–0.089* (0.051)	–0.058 (0.053)
Econ	–0.017 (0.035)	–0.057 (0.050)	–0.053 (0.051)
Science	–0.028 (0.038)	–0.071 (0.058)	–0.167*** (0.057)
Law	–0.141** (0.055)	–0.133 (0.090)	–0.082 (0.086)
Ivy League alumnus	–0.072* (0.042)	0.082 (0.058)	0.001 (0.059)
Generalist index	0.051*** (0.019)	–0.003 (0.026)	0.008 (0.026)
Recession graduate	–0.007 (0.034)	0.042 (0.048)	0.014 (0.053)
Fast track CEO	0.004 (0.004)	0.008* (0.005)	0.004 (0.005)
Obs	1017	936	936
Industry dummies	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
R-squared	0.114	0.279	0.286

Interpretation of this finding is not straightforward. The measure of financial expertise might be correlated

with omitted explanatory variables, which would bias the findings. We first address the concern of omitted variables

on CEO level in Section 5.1. We then discuss the concern that matching of CEOs to firms is endogenous, rather than random in Section 5.2.

5.1. Unobserved CEO heterogeneity

Financial expertise of CEOs might be correlated with other omitted variables on CEO-level. For instance, financial expert CEOs might have a different education or different general management skills that drive our findings. One advantage of our data set is that we observe detailed biographical information on the CEOs, including their educational background and employment history. This allows us to include additional controls. We add age, tenure, gender, a set of education dummies, controlling for educational background (MBA, economics, science, law), a generalist index controlling for the generic skills of the CEO, as well as some proxies for the talent of the CEO (Ivy League alumnus, recession graduate, fast track CEO). In all our tests in Tables 3 to 8 we control for CEO characteristics, alleviating the concern that financial expertise could be proxying for these. A concern that remains is that financial expertise might be correlated with CEO characteristics that are unobservable.

5.2. Endogenous CEO-firm matching

A second concern is that endogenous matching between CEOs and firms is driving our results. For instance, it could be the case that there are two types of firms, (1) low cash holding firms and (2) high cash holding firms, and that assortative matching between financial experts and low cash holdings firms based on preferences, for instance, is explaining our findings. In our primary analyses we rely on within-firm variation. This is a valid way to deal with the endogeneity concern if one believes that the matching between CEOs and firms is based on time-invariant unobservable characteristics. Under this assumption, the estimates of financial expertise can be interpreted as causal effects, and CEOs impose their particular style on a company. If optimal strategies of firms change over time, however, CEOs might not impose their styles but rather be chosen by firms because of their attributes.²¹

We show in Table 2 Panel C the average firm characteristics for firms with and without a financial expert CEO. Firms with financial expert CEOs are about 5.5 years older, on average, and they are almost double the size of firms without a financial expert CEO, \$5.9 billion and \$3.4 billion, respectively. The financial expertise of the CEO is also linked to less R&D (50% less investment in R&D), low-tech firms (30% fewer financial expert CEOs in high-tech firms), and more retained earnings (45% vs. 30%). Financial expert CEOs tend to run more diversified firms (25% more financial expert CEOs in diversified firms). Table 2 Panel C also shows the correlation between firm characteristics and the financial expertise of the CEO.

We conjecture that the demand for financial expertise in a CEO could be linked to a firm's life cycle, as we find that that firms run by financial experts tend to be older and larger, invest less in R&D, and have lower asset growth, lower cash flow volatility, and more retained earnings.²² All these characteristics are associated with firms that are at a mature stage in their life cycle.²³ Using the balance sheet as a metaphor, a firm in the growth stage of its life cycle is relatively more focused on the asset side, evaluating growth opportunities and projects, investing, and growing its assets. In a more mature stage the firm turns its focus to the financing side of the balance sheet, minimizing the cost of capital and paying out to shareholders while making sure it has enough financial resources to run its operations. A different type of CEO might be needed for firms at different stages of their life cycle. We conjecture that a financial expert CEO could be optimal for firms in a more mature stage, while a more entrepreneurial CEO could be better for firms in a growth stage. Note that this conjecture does not imply that financial expert CEOs are not valuable for growth firms as well, it implies that other skills might be relatively more important.²⁴ This hypothesis assumes a production function that allows for complementarities between CEO skills and other firm resources.²⁵

In Table 9 we analyze matching CEOs and firms employing a multivariate setup. To test whether firms that hire financial expert CEOs and firms that hire nonfinancial expert CEOs are different, we analyze firm characteristics at the time they hire a new CEO. In specifications (1)–(10), we restrict the sample to the last year of tenure of a CEO; i.e., we look at years that are followed by a CEO turnover. In specifications (11) and (12) we are even more restrictive and make within-firm comparisons, including only firms with at least two CEO turnovers of different types: in one the incoming CEO is a financial expert CEO, and in one the CEO is a nonfinancial expert.²⁶ The dependent variable in specifications (1)–(12) is a dummy variable that is equal to one if the new CEO is a financial expert and zero otherwise. We find results consistent with the univariate analysis. Firms that are larger, have fewer investment opportunities, and less volatile assets, are older, and have more retained earnings are more likely to have a financial expert CEO. All these characteristics are typical of firms that are not in a growth stage of their life cycle. In columns 9 and 10, we include all the characteristics simultaneously. As many of these variables are correlated, in these

²² The life cycle theory of the firm goes back to Mueller (1972), who refers to an S-shaped growth pattern of slow growth at start-up, rapid growth at the expansion or growth stage, and slow growth when maturing.

²³ See also Bulan and Subramanian (2009), DeAngelo, DeAngelo, and Stulz (2006), and Fama and French (2001).

²⁴ To make the argument work, we need to assume that financial expertise comes at the expense of some other skill. We provide evidence for this claim in Sections 5.2.4, 5.2.5, and 6.

²⁵ Note that it is plausible that firms “reinvent” themselves and alternate between stages of growth and maturity. This can happen through entering new product markets or product innovation, and would also be consistent with our results.

²⁶ As turnovers are relatively rare, a fixed effects estimator is not feasible in this setup as that requires a number of observations per group.

²¹ See also Bertrand and Schoar (2003) or Fee, Hadlock, and Pierce (2013).

specifications only size, total investment, and asset growth are significant.

Overall, these results support the idea of an assortative match of CEOs with different levels of financial expertise to firms at different stages of their life cycle. When we restrict the analysis to firms that have at least two turn-overs of different types, we lose about two-thirds of the observations, and the tests have less power. The point estimates, however, remain the same in terms of sign and magnitude, as compared to the previous tests. The coefficient on retained earnings remains insignificant.

In specifications (13) and (14) we include all firm-years. The dependent variable is a dummy that is equal to one if the current CEO is a financial expert. We find that financial expert CEOs are, on average, employed at firms that are larger, have lower total investments, and lower asset volatility, are older, and experience less asset growth.

The finding that financial expert CEOs tend to be matched to firms in more mature stages is important as random matching between CEOs and firms is an implicit identifying assumption in many research papers on CEO characteristics and their influence on firm performance. One main contribution of this paper is to provide direct evidence on the matching between firms and CEOs based on both observable CEO and observable firm characteristics. Little attention has been paid in the literature to matching itself. We do not know whether or why certain types of CEOs are hired in specific situations or, more generally, the sources (if any) of match-specific productivity (see Oyer and Schaefer, 2011).

In our case, the findings are consistent with two non-mutually exclusive views. Under the “financial skills view,” financial expertise is relevant to implement a given financial policy, and firms optimally hire financial experts at the life cycle stage when financial policies are more important than other firm policies. The results are also consistent with a view under which (skills of) CEOs do not matter (“no financial skills view”). For instance, financial expert CEOs could have an idiosyncratic preference for specific financial policies, and firms hire CEOs with the preferences to match their life cycle stage or their business model.²⁷ The observed assortative matching is based on the conjecture that financial expert CEOs have a preference for low cash holdings, high leverage, and high payouts and would match themselves to firms accordingly. In this case, appointing a financial expert CEO is the *cheapest* way to implement a certain strategy. There can also exist other reasons, not directly associated with financial expertise, for why we observe an assortative matching between financial expert CEOs and mature firms. For instance, these CEOs might have some useful primary skill set that tends to be correlated with financial expertise, i.e., observing financial expertise is a side-effect of this primary skill and unrelated to preferences.

²⁷ The matching has to be dependent on the life cycle of the firm, as our results are robust to relying only on within-firm variation, i.e., matching solely on time-invariant firm characteristics cannot explain our results. This is in contrast to Cronqvist, Makhija, and Yonker (2012) who show that firms and CEOs match according to their leverage preferences.

To distinguish convincingly between these two views, it would be ideal to allocate CEOs with different sets of skills randomly to firms at different stages in their life cycle, e.g., allocate a nonfinancial expert CEO to a mature firm. This ideal experiment would deal with breaking the optimal and endogenous matching. In our tests, however, to estimate any effect of financial expertise, we need to assume that there are some frictions in the matching between CEOs and firms; i.e., the matching is not optimal at all times. We believe that this assumption is not implausible. As we observe the labor markets for executives, we do not find firms changing CEOs that often. If there are some costs (such as search costs) or there is valuable firm-specific knowledge, firms might not want to change CEOs too frequently.

In the previous sections, we have already shown some evidence that is mostly consistent with the “financial skills view.” We have direct evidence that financial expert CEOs are more sophisticated and follow finance theory more closely. Moreover, we exploit exogenous shocks to overall credit conditions and show that financial expert CEOs are more able to access credit markets in bad times. If one believes that these shocks were not anticipated, and hence the matching between CEOs and firms is assumed to be exogenous with respect to their occurrence, these tests increase our confidence in a “financial skills view” interpretation. We also present further evidence that is mainly consistent with a “financial skills view” interpretation of our findings in the following sections.

5.2.1. Controlling for the firm life cycle

If we believe that the matching between financial expert CEOs and firms relates to the life cycle of the firm, controlling for the life cycle will help us estimate the causal effect of financial expertise. Indeed, this strategy allows us to exploit variation within the life cycle of a firm by estimating the difference in cash holdings of mature firms that are run by financial expert CEOs and nonfinancial expert for instance. We thus add firm size, asset volatility, firm age, total investment, asset growth, and retained earnings as proxies for the life cycle to our standard specifications. As it is plausible that the life cycles of firms within the same industry are correlated, we include industry interacted with year dummies in our regressions.

Adding these controls attempts to address the possibility that financial expertise might be capturing only the effects of firm life cycle, given that these CEOs tend to be matched to more mature firms. The identifying assumptions are that (1) the matching between firms and financial expert CEOs is based on these characteristics, and (2) the matching between CEOs and firms is not perfect all the time.

Table 10 reports results that are very similar in terms of magnitude and statistical significance as the findings in Table 3, which supports a “financial skills view” interpretation of our findings.

5.2.2. Exogenous shocks

Even though we do not observe exogenous variation in our main variable of interest, the financial expertise of

Table 9

Specifications 1–10 analyze turnovers only and the sample is restricted to firm-year observations of turnover years, i.e., includes only firm-year before replacement of CEO in the next year. Specifications 11 and 12 analyze a subsample of these turnovers. Only firms with at least two different turnovers (i.e., the firm is hiring a financial expert in one and a nonfinancial expert in the other turnover) are included. The dependent variable in regressions 1–12 is a dummy equal to one if the incoming CEO is a financial expert. Specifications 13 and 14 analyze all firm-years. The dependent variable is a dummy variable that is equal to one if the current CEO is a financial expert CEO. Standard errors are clustered at firm level. Variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(9)	(10)	(11)	(12)	(13)	(14)
	Incoming CEO is financial expert conditional of a turnover in $t+1$											Financial expert	
Log(assets)	0.045*** (0.007)							0.039*** (0.009)	0.034*** (0.011)	0.008 (0.014)	0.001 (0.022)	0.047*** (0.003)	0.043*** (0.003)
Capex		−0.462*** (0.133)						−0.341** (0.160)	−0.369* (0.218)	−0.392 (0.315)	−0.083 (0.564)	−0.368*** (0.054)	−0.410*** (0.062)
Log q			−0.019 (0.018)					0.033 (0.021)	0.038 (0.029)	0.017 (0.038)	0.032 (0.072)	0.003 (0.015)	0.009 (0.016)
Asset volatility				−0.206*** (0.056)				−0.031 (0.076)	−0.025 (0.112)	−0.138 (0.129)	−0.111 (0.252)	−0.106*** (0.026)	−0.098*** (0.032)
Firm age					0.003*** (0.001)			0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.002)	0.002*** (0.000)	0.002*** (0.000)
Asset growth						−0.064 (0.039)		−0.091** (0.040)	−0.101** (0.051)	−0.186*** (0.069)	−0.215* (0.112)	−0.032** (0.013)	−0.033** (0.014)
Retained earnings							0.005 (0.007)	−0.006 (0.007)	−0.003 (0.009)	−0.017 (0.012)	−0.008 (0.019)	−0.003 (0.003)	−0.003 (0.003)
Obs	1871	1871	1871	1856	1858	1737	1865	1722	1722	650	650	15738	15738
Sample	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover	Turnover (two different ones)	Turnover (two different ones)	Full	Full
Industry × Year dummies	No	No	No	No	No	No	No	No	Yes	No	Yes	No	Yes
R-squared	0.023	0.005	0.001	0.007	0.011	0.001	0.000	0.029	0.313	0.020	0.463	0.049	0.107

Table 10

In these regressions we explicitly control for the life cycle of the firm by including size, asset volatility, firm age, total investment, asset growth, retained earnings, and by including industry $\times$ year dummies accounting for the fact that the life cycle of the industry is a proxy for the life cycle of the firm. The dependent variable in regression (1) is the log of cash and marketable securities. The dependent variable in regression (2) is the ratio of total debt to assets. The dependent variable in regression (3) is a dummy variable equal to one if the firm paid a dividend in the current year. The dependent variable in regression (4) is a dummy variable equal to 1 if the firm repurchased shares in the current year. Year dummies are included in all regressions. Variable definitions are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1) Cash	(2) Leverage	(3) Dividends	(4) Repurchases
Financial expert CEO	−0.104** (0.050)	0.015** (0.006)	0.004 (0.009)	0.037* (0.019)
Age	−0.066** (0.028)	0.003 (0.003)	0.003 (0.005)	−0.002 (0.011)
Age squared	0.001** (0.000)	−0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)
Sex	−0.211 (0.161)	−0.007 (0.017)	−0.025 (0.041)	−0.060 (0.055)
Tenure	−0.004 (0.004)	−0.001 (0.001)	0.001 (0.001)	0.002 (0.002)
Log(assets)	−0.567*** (0.041)	0.012* (0.006)	0.032*** (0.008)	0.050*** (0.019)
Log q	0.389*** (0.042)	−0.009 (0.007)	0.021*** (0.007)	−0.072*** (0.022)
Asset volatility	0.268*** (0.094)	−0.173*** (0.016)	−0.043** (0.022)	−0.131** (0.051)
Dividend dummy	0.142** (0.056)	−0.023*** (0.008)		
ROA		−0.241*** (0.039)	0.055 (0.044)	0.724*** (0.108)
Leverage	0.323*** (0.084)		−0.093*** (0.027)	−0.272*** (0.061)
Cash flow	0.285*** (0.083)			
Net working capital	−0.401*** (0.142)			
Acquisitions	−1.306*** (0.149)			
Dividend dummy ($t-1$)			0.593*** (0.019)	0.018 (0.026)
Cash			0.022** (0.009)	0.053* (0.028)
MBA	0.016 (0.055)	0.001 (0.008)	−0.006 (0.010)	0.048** (0.023)
Econ	−0.038 (0.056)	0.006 (0.007)	−0.006 (0.010)	−0.027 (0.023)
Science	−0.077 (0.061)	0.001 (0.008)	−0.002 (0.011)	−0.033 (0.026)
Law	−0.077 (0.084)	0.015 (0.010)	−0.029 (0.018)	0.004 (0.045)
Ivy League alumnus	−0.011 (0.063)	0.006 (0.008)	−0.013 (0.010)	−0.031 (0.025)
Generalist index	0.051* (0.027)	−0.000 (0.003)	−0.003 (0.005)	0.011 (0.012)
Recession graduate	−0.010 (0.053)	−0.000 (0.006)	−0.006 (0.009)	0.007 (0.022)
Fast track CEO	−0.002 (0.003)	0.000 (0.000)	0.000 (0.001)	0.001 (0.001)
Firm age	0.008 (0.005)	−0.000 (0.001)	0.002** (0.001)	0.003 (0.002)
Total investments	−2.563*** (0.268)	−0.197*** (0.061)	0.067 (0.048)	0.070 (0.132)
Asset growth	0.168*** (0.037)	0.030*** (0.006)	−0.004 (0.008)	−0.078*** (0.019)
Retained earnings	0.006 (0.009)	−0.001 (0.003)	0.001 (0.002)	0.007 (0.004)
Obs	12423	12493	11891	11891
Industry $\times$ Year dummies	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes

Table 11

while restricting the sample to firms headed by CEOs who are succeeding a CEO who reached retirement age. The specifications are the same as in Table 2. CEO and firm variable are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1) Cash	(2) Leverage	(3) Dividends	(4) Repurchases
Financial expert CEO	−0.205** (0.101)	0.030** (0.013)	0.009 (0.016)	0.049 (0.036)
Age	−0.087** (0.043)	0.005 (0.006)	−0.001 (0.008)	−0.020 (0.019)
Age squared	0.001** (0.000)	−0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Sex	−0.508** (0.253)	−0.028 (0.041)	−0.086 (0.095)	−0.105 (0.120)
Tenure	−0.002 (0.007)	−0.002* (0.001)	−0.000 (0.001)	0.002 (0.003)
Log(assets)	−0.584*** (0.093)	0.023 (0.016)	0.032* (0.017)	0.089** (0.041)
Log q	0.375*** (0.095)	−0.014 (0.015)		
Asset volatility	0.135 (0.237)	−0.181*** (0.039)		
R&D	0.708 (0.864)	−0.104 (0.153)		
Capex	−4.017*** (1.007)	−0.342*** (0.112)		
Dividend dummy	0.057 (0.119)	−0.012 (0.017)		
ROA		−0.192** (0.089)	0.026 (0.082)	0.798*** (0.274)
Leverage	0.204 (0.210)			
Cash flow	0.482** (0.207)			
Net working capital	−0.527 (0.361)			
Acquisitions	−1.376*** (0.365)			
Dividend dummy($t-1$)			0.698*** (0.039)	0.048 (0.067)
Leverage			−0.054 (0.053)	−0.122 (0.137)
Cash			0.004 (0.020)	0.002 (0.064)
MBA	0.063 (0.102)	0.017 (0.017)	0.011 (0.027)	0.037 (0.047)
Econ	−0.011 (0.110)	0.004 (0.014)	−0.000 (0.019)	0.006 (0.044)
Science	0.022 (0.122)	0.019 (0.019)	−0.021 (0.020)	−0.030 (0.058)
Law	0.022 (0.157)	0.005 (0.017)	−0.031 (0.031)	−0.067 (0.093)
Ivy League alumnus	−0.094 (0.099)	0.011 (0.014)	−0.027 (0.025)	−0.063 (0.050)
Generalist index	0.036 (0.057)	−0.013* (0.007)	−0.002 (0.010)	−0.005 (0.024)
Recession graduate	−0.001 (0.114)	0.005 (0.014)	0.016 (0.019)	0.014 (0.046)
Fast track CEO	0.002 (0.005)	0.000 (0.001)	−0.001 (0.001)	0.001 (0.002)
Firm age	0.014 (0.011)	0.000 (0.001)	0.002 (0.001)	−0.000 (0.003)
Asset growth	0.227** (0.094)	0.017 (0.018)	−0.002 (0.020)	−0.084* (0.045)
Retained earnings	0.017 (0.025)	−0.003 (0.006)	0.002 (0.006)	0.005 (0.012)
Obs	2751	2768	2616	2616
Life cycle controls	Yes	Yes	Yes	Yes
Industry × Year dummies	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Adjusted R-squared	0.790	0.698	0.893	0.392

Table 12

Columns 1 and 2 show the effect of firm age and financial expertise of the CEO on firm performance. The dependent variable in all regressions is the log of q . Columns 3 and 4 present estimates of OLS and firm fixed effects panel regressions of the logarithm of CEO total and equity pay on the financial expertise of the CEO- and firm-level control variables. CEO and firm variables are as defined in Table 1. Standard errors clustered at firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)
	Tobin's q		Total pay	Equity pay
Financial expert CEO $\times$ Asset growth	–0.122*** (0.040)			
Financial expert CEO $\times$ Firm age		0.002** (0.001)		
Financial expert CEO	0.007 (0.017)	–0.056* (0.031)	–0.040 (0.035)	0.002 (0.056)
Firm age	0.002** (0.001)	0.001 (0.001)	0.016*** (0.005)	0.014** (0.006)
Asset growth	0.374*** (0.027)			
Log(assets)	–0.299*** (0.021)	–0.293*** (0.020)		
Leverage	–0.465*** (0.061)	–0.440*** (0.064)		
PPE	–0.487*** (0.106)	–0.875*** (0.115)		
Capex	1.317*** (0.158)	1.808*** (0.174)		
R&D	0.176 (0.438)	–0.981** (0.490)		
Age	0.003 (0.010)	–0.014 (0.011)	0.028 (0.025)	–0.009 (0.031)
Age squared	–0.000 (0.000)	0.000 (0.000)	–0.000 (0.000)	0.000 (0.000)
Sex	–0.002 (0.051)	0.017 (0.055)	–0.015 (0.098)	–0.390** (0.189)
Tenure	0.003** (0.001)	0.004*** (0.001)	–0.005 (0.003)	–0.009* (0.005)
Asset volatility			0.423*** (0.086)	0.938*** (0.100)
External hired CEO			0.232*** (0.037)	0.347*** (0.062)
First year CEO			0.157*** (0.035)	0.272*** (0.055)
CEO-chairman dummy			0.020 (0.030)	0.044 (0.047)
Log market value of assets			0.565*** (0.031)	0.776*** (0.041)
Market to book			–0.057*** (0.018)	–0.030 (0.022)
ROA			0.681*** (0.193)	0.090 (0.253)
ROA($t-1$)			–0.024 (0.167)	0.365 (0.232)
Stock returns			–0.018 (0.021)	–0.262*** (0.030)
Stock returns ($t-1$)			0.094*** (0.017)	0.057** (0.025)
Obs	14782	16317	10929	7423
Year dummies	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Adjusted R -squared	0.730	0.688	0.673	0.686

CEOs, we are able to observe how the two types of CEOs react to exogenous changes in the business environment. The choice of a CEO is endogenous; however, firms are unlikely to frequently change their CEOs to systematically respond to changes in the environment. We make use of exogenous changes in credit market conditions and

changes to the tax code (see Sections 4.4. and 4.5) to test whether financial experts behave differently from non-experts. We find consistent evidence that the two types of CEOs respond differently to these changes. Moreover, we find that financial expert CEOs respond in a way that is consistent with them making use of their financial skills.

Table 13

The dependent variable in column 1 is the log of cash and marketable securities. The dependent variable in column 2 is the ratio of total debt to assets. The dependent variable in column 3 is a dummy variable equal to one if the firm paid a dividend in the current year. The dependent variable in column 4 is a dummy variable equal to one if the firm repurchased shares in the current year. We use the nearest-neighbor estimator (nnmatch) from [Abadie, Drukker, Leber Herr, and Imbens \(2004\)](#). We use the same set of firm variables as in our baseline regressions ([Table 4](#)) for matching. Moreover, we require that the treated and the control observation(s) match exactly on industry (two-digit SIC) and year. Variable definitions are as defined in [Table 1](#). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Cash		Leverage		Dividends		Repurchases	
Financial expert CEO	−0.204*** (0.024)	−0.219*** (0.023)	0.005* (0.003)	0.008*** (0.003)	0.016*** (0.004)	0.034*** (0.005)	0.045*** (0.009)	0.045*** (0.008)
Obs	15,643	15,643	15,720	15,720	14,950	14,950	14,950	14,950
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exact matching	Industry, Year	Industry, Year	Industry, Year	Industry, Year	Industry, Year	Industry, Year	Industry, Year	Industry, Year
# Of neighbors	1	4	1	4	1	4	1	4

5.2.3. “Exogenous” turnovers: evidence from retiring CEOs

We also analyze a subsample of firms where CEO turnover is likely to occur for exogenous reasons and is less likely to be related to the life cycle of the firm. We classify turnovers where the incumbent CEO retires—is older than 65 years—as “more” exogenous, following [Jenter and Lewellen \(2011\)](#). The idea is that right before the retirement of a CEO, firms might not have what would be the optimal CEO in terms of financial expertise. Consider that because the CEO is due to retire soon, the firm might have delayed replacing a nonfinancial expert with a financial expert. This would help us to identify the effect of financial expertise.

[Table 11](#) shows the results. When we restrict our sample to turnovers after retirements, the panel is reduced to 2,768 observations. Consistent with the previous findings, the effect of financial expertise is significant for all financial policies: cash, leverage, and payout. We find that when the newly appointed CEO is a financial expert, firms tend to hold less cash and more debt, pay more dividends, and engage more in share repurchases.

One caveat of this approach is that even though the turnover occurs for exogenous reasons, the choice of the new CEO is likely not random. The argument is rather that because the turnover occurs for exogenous reasons, the timing of CEO replacement is likely not to be optimal. Therefore, if financial expertise is relevant for implementing a given financial policy, we should observe an effect of switching to a financial expert after a retirement.

5.2.4. CEO-Firm matching and firm performance

Next, we look at the valuation effects of different CEOs at different stages of a firm's life cycle. Under the “no financial skills view,” the identity of the CEO is not expected to matter. Under the “financial skills view,” however, there might be valuation implications if there are some frictions in the labor markets for CEOs. Assuming that it is beneficial for a more mature firm to appoint a financial expert CEO, if some of the more mature firms do not do it (or cannot do it), we should see a positive effect on performance of interaction between the financial

expertise of the CEO and mature firms. Columns 1 and 2 of [Table 12](#) show the results of the tests. The dependent variable in all regressions is q . The independent variable of interest is the interaction between the financial expert dummy and asset growth (in specification 1) and between the financial expert dummy and firm age (specification 2).

We find a negative effect for the interaction between asset growth and financial expertise. This result suggests that firm resources in mature firms are complements to the financial expertise of the CEO, consistent with the “financial skills view” and do not support the “no financial skills view” hypothesis. Consistently, we find a positive and significant coefficient for the interaction of firm age with the financial expertise of the CEO when we use age as a proxy for maturity of a firm.

5.2.5. Are financial expert CEOs less costly?

As discussed before, under the “no financial skills” hypothesis, one potential explanation for the assortative matching and the differences in financial policies are purely CEO preferences. One direct test of an explanation based on preferences is to look at compensation. For instance, a financial expert who prefers to be employed in high-leverage firms might accept a lower salary than a nonfinancial expert. In this case, hiring a financial expert CEO might be the cheapest way to implement a high-leverage strategy. We test if this is the case by running compensation regressions. We look at total pay and equity compensation. Columns 3 and 4 of [Table 12](#) show the results.

We find no evidence that financial expert CEOs are paid at a discount. Column 3 shows the regressions of total compensation, including cash and equity compensation on the financial expert dummy and a set of firm-level and CEO-level controls, as commonly used in the CEO compensation literature (see, for instance, [Custódio, Ferreira, and Matos, 2013](#)). The point estimate for financial expertise is actually negative, although not statistically different from zero. Column 4 shows the results for equity compensation. Again, there is no significant difference between financial expert and nonfinancial expert CEOs.

The point estimate is very low and statistically not different from zero.

Another explanation for our results on financial policies might be that financial expert CEOs are different in terms of risk tolerance. Being more risk tolerant could be one reason why they hold less cash and more debt. The results for compensation seem to reject this hypothesis, because we observe no differences in terms of pay mix among financial and nonfinancial experts. If indeed financial experts value risk differently, we would expect differences in terms of their cash vs. equity pay mix.

5.2.6. Matching estimator

We use a matching estimator as an alternative method to estimate the effect of financial expertise on financial policies. We are aware of the fact that matching estimator does not mitigate the concern of endogenous CEO-firm matching. A causal interpretation of the matching estimate relies on the same assumptions as the estimated coefficient in a regression model. The matching estimator, however, allows us to match certain observable characteristics exactly (which could be important in our case). Keeping in mind that the life cycles of firms within the same industry are potentially correlated, we require exact matching based on industry (SIC two-digit) and year. We use proxies for the life cycle of the firm, CEO background variables, and further firm controls to calculate propensity scores. We then estimate the average treatment of the treated effect using nearest-neighbor matching estimator with replacement from [Abadie, Drukker, Leber Herr, and Imbens \(2004\)](#). [Table 13](#) shows the results. We find statistically significant effects of a similar magnitude as in our previous regression tests.

6. Financial expert CEOs and nonfinancial corporate policies

Financial expertise might come at the expense of having less expertise or skills in other dimensions. Indeed, the findings on valuation and compensation do not indicate that financial expert CEOs are generally superior. The results are consistent with the idea that there exist complementarities between different skills of the CEO and firm characteristics: e.g., financial expert CEOs seem to have skills that are valuable in mature companies, i.e., companies that are at a later stage in their life cycle. If, on the other hand, financial CEOs lack skills that are likely to be important at other stages of the lifetime of a corporation, they are likely to underperform in these companies—for instance young firms. Columns 1 and 2 of [Table 12](#) in the paper show results that are consistent with this hypothesis. The interaction effects between financial expertise and different proxies for the life cycle of the firm (age and asset growth) have signs accordingly. In other words, we find that financial CEOs underperform nonfinancial experts in young firms or high-growth firms (proxies for an early stage in the life cycle), while they outperform nonfinancial expert CEOs in more mature companies. For instance, a financial expert in a young firm

(lowest age quartile) underperforms a nonfinancial expert in a young firm by 4% in terms of Tobin's q . Similarly, a financial expert CEO in a high-growth firm (top quartile) underperforms a nonfinancial expert in a high-growth firm by 2%.

While these findings are indirect evidence that financial expert CEOs lack other relevant skills to perform their role, we also analyze nonfinancial policies where financial expertise could be less or not important (and financial expert CEOs could lack expertise).

6.1. Innovation

First, we analyze whether financial expert CEOs are worse innovators. We use the National Bureau of Economic Research (NBER) patent database (2006 edition) to measure innovation ([Hall, Jaffe, and Trajtenberg, 2001, 2005](#)).²⁸ We control for firm characteristics as in previous tests, and for institutional ownership using data from the Thomson CDA/Spectrum 13 F Holdings (see [Aghion, Van Reenen, and Zingales, 2013](#)).

The first measure of innovation we use in our tests is R&D spending. We use the ratio of R&D expenditures to the book value of assets (R&D). Firm-years with missing R&D information are assigned a zero value.

The second measure of innovation is the number of patent applications filed by a firm in a given year (Patents). On average, there is a two-year lag between patent application and grant date. Since the latest year available in the patent database is 2006, patents applied for in 2004 and 2005 could not show up in the database. We end our sample period in 2003 and include year fixed effects in our regressions to address time-truncation issues ([Hall, Jaffe, and Trajtenberg, 2001](#)).

The third measure of innovation is the total number of citations received by the set of patents that a firm applied for in a given year (Citations). A typical way to measure the relevance of a patent is its citation count which takes into account both the number of patents and the number of citations subsequently received from other patents ([Griliches, Hall, and Pakes, 1987; Hall, Jaffe, and Trajtenberg, 2005](#)). However, more recent patents have less time to accumulate citations, and for this reason citations suffer from a time-truncation bias. We address this concern by adjusting the citation count of each patent. The adjustment consists of multiplying each patent's citation count by the weighting index from [Hall, Jaffe, and Trajtenberg \(2001, 2005\)](#). The resulting variable, Citations Q , is the sum of the adjusted patent citations across all patents applied for in each year.

Following [Aghion, Van Reenen, and Zingales \(2013\)](#), we introduce “firm fixed effects” into the models using the “pre-sample mean scaling” method first proposed by [Blundell, Griffith, and Van Reenen \(1999\)](#). We exploit the fact that we have a long pre-sample history on R&D expenditures and patenting behavior (of up to 25 years per firm) to construct the pre-sample average of citation-weighted patents. This can then be used as an initial

²⁸ The NBER patent database provides a link to Execucomp by GVKEY.

Table 14

The dependent variable in columns 1 and 2 is R&D spending. The dependent variable in columns 3–5 is the number of patent applications filed by a company in a given year. The dependent variable in columns 6 through 8 is the total number of citations received by the set of patents that a firm applied for in a given year. The dependent variable in columns 9 through 11 is the number of citations (as before) adjusted for time-truncation as in Hall, Jaffe, and Trajtenberg (2001, 2005). We estimate specifications 1 and 2 by OLS. In specifications 3 to 11 we use negative binomial regressions and introduce pre-sample mean scaling as in Aghion, Van Reenen, and Zingales (2013) and Blundell, Griffith, and Van Reenen (1999). Variable definitions are as defined in Table 1. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	R&D		log Patents			log Citations			log Citations (Q)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Financial expert CEO	−0.559*** (0.202)	−0.816*** (0.246)	−0.007 (0.045)	−0.130** (0.062)	−0.128* (0.071)	−0.029 (0.053)	−0.152** (0.069)	−0.181** (0.079)	−0.014 (0.057)	−0.157** (0.071)	−0.190** (0.081)
Financial expert CEO × R&D					0.016* (0.009)			0.024** (0.011)			0.027** (0.012)
Age	0.052 (0.124)	−0.215 (0.141)	0.012 (0.030)	−0.036 (0.036)	−0.010 (0.035)	0.008 (0.033)	−0.040 (0.042)	−0.020 (0.041)	0.005 (0.035)	−0.042 (0.043)	−0.023 (0.042)
Age squared	−0.001 (0.001)	0.001 (0.001)	−0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Gender	0.555 (0.829)	1.714* (0.966)	0.026 (0.148)	0.430** (0.213)	0.262 (0.214)	0.053 (0.197)	0.654*** (0.241)	0.374 (0.263)	0.048 (0.194)	0.644** (0.276)	0.354 (0.271)
Tenure	−0.010 (0.015)	0.008 (0.024)	−0.000 (0.004)	0.001 (0.007)	−0.004 (0.006)	−0.004 (0.005)	−0.004 (0.008)	−0.007 (0.007)	−0.005 (0.005)	−0.005 (0.008)	−0.007 (0.007)
R&D					0.080*** (0.006)			0.088*** (0.008)			0.093*** (0.009)
Log(sales)	−0.831*** (0.120)	−1.209*** (0.141)	0.340*** (0.020)	0.119*** (0.028)	0.237*** (0.024)	0.316*** (0.024)	0.096*** (0.032)	0.219*** (0.029)	0.290*** (0.023)	0.071** (0.032)	0.193*** (0.030)
Log(PPE)	0.164 (0.196)	−0.743*** (0.129)	0.197*** (0.047)	−0.010 (0.035)	0.061* (0.032)	0.168*** (0.050)	−0.068* (0.040)	0.009 (0.037)	0.156*** (0.047)	−0.089** (0.042)	−0.009 (0.039)
Stock returns	0.188* (0.101)	0.286** (0.122)	0.065*** (0.020)	0.075*** (0.028)	0.024 (0.023)	0.043* (0.026)	0.056 (0.034)	0.007 (0.031)	0.030 (0.027)	0.037 (0.036)	−0.012 (0.033)
IOR	−1.397** (0.556)	−0.487 (0.635)	−0.105 (0.149)	0.148 (0.153)	0.312** (0.145)	−0.031 (0.172)	0.201 (0.176)	0.358** (0.172)	−0.049 (0.174)	0.191 (0.179)	0.324* (0.175)
MBA		0.016 (0.286)		0.136* (0.070)	0.129** (0.060)		0.173** (0.082)	0.161** (0.070)		0.189** (0.084)	0.174** (0.072)
Econ		−0.004 (0.263)		0.054 (0.075)	0.091 (0.066)		0.077 (0.080)	0.122* (0.070)		0.079 (0.082)	0.119* (0.072)
Science		2.997*** (0.330)		0.413*** (0.068)	0.208*** (0.060)		0.517*** (0.077)	0.302*** (0.070)		0.557*** (0.078)	0.339*** (0.074)
Law		0.142 (0.475)		−0.091 (0.153)	−0.197 (0.130)		−0.098 (0.160)	−0.232* (0.135)		−0.136 (0.160)	−0.268** (0.132)
Ivy league alumnus		0.775** (0.355)		−0.025 (0.074)	−0.063 (0.064)		−0.077 (0.083)	−0.128* (0.075)		−0.058 (0.086)	−0.107 (0.081)
Generalist index		0.649*** (0.132)		0.111*** (0.035)	0.057* (0.030)		0.131*** (0.039)	0.067* (0.035)		0.160*** (0.039)	0.083** (0.036)
Recession graduate		0.242 (0.285)		0.144** (0.065)	0.123** (0.059)		0.149** (0.071)	0.147** (0.067)		0.163** (0.073)	0.172** (0.071)
Fast track CEO		0.027 (0.019)		0.003 (0.006)	−0.001 (0.005)		0.002 (0.006)	−0.001 (0.005)		0.002 (0.006)	−0.001 (0.005)
Obs	8042	7601	8042	7601	7601	8042	7601	7601	8042	7601	7601
Industry dummies	Yes	No	Yes	No	No	Yes	No	No	Yes	No	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-sample mean scaling	No	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
R squared/Pseudo R squared	0.431	0.249	0.126	0.078	0.122	0.093	0.062	0.086	0.060	0.037	0.054

condition to proxy for unobserved heterogeneity under certain conditions (in particular, the first moments of the variables must be stationary).

Table 14 shows the results. In columns 1 and 2 we look at R&D, 3–5 at patents, and 6–11 at citations. The R&D results using OLS estimates suggest that firms with financial experts tend to spend less in R&D. The estimate is economically significant: compared to the mean, financial expert CEOs spend 25% less in R&D. We further analyze this issue by looking at the number of filed patents and the number of citations. We find that financial expert CEOs file up to 13% fewer patents than non-experts, and generate up

to 19% fewer citations. We find similar results when using a citation measure that corrects for time-truncation bias.²⁹

In columns 5, 8 and 11 we look at the sensitivity of innovation output measures to R&D spending. The coefficient of interest in these regressions is the

²⁹ The results are also robust to an alternative citation measure that corrects for the time-truncation bias. This second adjustment consists of scaling each patent's citation count by the average citation count of all patents in the same two-digit technology class and year. The resulting variable, TT Citations, is the sum of the adjusted citation count across all patents applied for in each year.

Table 15

The dependent variable in column 1 is the logarithm of advertising spending. The dependent variable in columns 2 and 3 is the logarithm of next year sales. The dependent variable in columns 4 and 5 are total wage expenses divided by the number of employees. Variable definitions are as defined in Table 1. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Log advertising	Log sales ($t+1$)			Average wage	
	(1)	(2)	(3)	(4)	(5)	(6)
Advertising $\times$ Financial expertise		−0.040*** (0.015)	−0.010 (0.011)	−0.009 (0.010)		
Log advertising		0.129*** (0.014)	0.116*** (0.012)	0.045** (0.017)		
Financial expert CEO	0.009 (0.078)	0.060 (0.055)	0.028 (0.048)	0.048 (0.042)	−1.190 (1.965)	−0.385 (1.991)
Log(assets)	0.939*** (0.030)	0.761*** (0.018)	0.761*** (0.016)	0.580*** (0.027)	1.623 (1.025)	0.451 (2.430)
Log q	0.205*** (0.068)	0.244*** (0.033)	0.235*** (0.025)	0.207*** (0.022)	−3.935 (2.647)	−1.547 (1.897)
Firm age	0.005* (0.003)	0.002* (0.001)	0.001 (0.001)	0.005* (0.003)	−0.069 (0.076)	0.127 (0.088)
Cash flow	0.214 (0.209)	0.242*** (0.081)	0.223*** (0.074)	0.167** (0.083)	5.393 (7.624)	0.164 (5.694)
Net working capital	0.016 (0.219)	0.263*** (0.094)	−0.127 (0.104)	0.012 (0.065)	−44.064*** (13.068)	−15.487 (9.558)
Capex	1.533** (0.723)	1.024*** (0.353)	0.397 (0.272)	−0.027 (0.168)	−28.904** (14.366)	2.706 (10.934)
Asset growth	−0.497*** (0.075)	−0.122*** (0.034)	−0.102*** (0.029)	−0.078*** (0.019)	−6.669*** (2.191)	−5.900*** (2.062)
Stock return		0.090*** (0.016)	0.078*** (0.013)	0.032*** (0.011)		0.616 (0.762)
Asset volatility		−0.036 (0.099)	−0.150* (0.086)	−0.152*** (0.050)		7.066 (4.689)
R&D		−1.422*** (0.197)	−1.249*** (0.195)	−0.641** (0.261)		39.633 (57.293)
Retained earnings		−0.004 (0.007)	−0.004 (0.006)	0.004 (0.005)		0.948 (0.970)
Age		0.017 (0.032)	−0.004 (0.017)	−0.000 (0.014)		−1.481 (1.227)
Age squared		−0.000 (0.000)	0.000 (0.000)	0.000 (0.000)		0.011 (0.011)
Gender		0.095 (0.101)	0.172** (0.077)	0.017 (0.033)		2.716 (3.807)
Tenure		−0.005 (0.004)	−0.004 (0.003)	0.000 (0.002)		0.181 (0.186)
MBA		−0.058 (0.037)	−0.026 (0.026)	0.022 (0.030)		−3.720* (2.121)
Econ		0.026 (0.038)	0.018 (0.030)	−0.022 (0.025)		0.151 (2.458)
Science		−0.037 (0.040)	−0.006 (0.034)	0.018 (0.024)		2.910 (3.676)
Law		−0.051 (0.079)	−0.035 (0.052)	−0.032 (0.047)		−4.022 (4.183)
Ivy League alumnus		−0.138*** (0.044)	−0.081*** (0.031)	−0.056 (0.036)		1.932 (2.245)
Generalist index		−0.009 (0.016)	−0.009 (0.013)	0.006 (0.012)		0.106 (1.305)
Recession graduate		−0.009 (0.036)	−0.025 (0.029)	0.036 (0.026)		−0.274 (2.153)
Fast track CEO		−0.005 (0.003)	−0.003 (0.002)	−0.000 (0.001)		0.097 (0.089)
Obs	5398	3473	3473	3473	1412	1068
Industry dummies	No	No	Yes	No	Yes	No
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	No	No	Yes	No	Yes
Adjusted R-squared	0.608	0.916	0.947	0.989	0.607	0.925

interaction term between R&D spending and financial expertise. We find this coefficient to be positive and significant across the different measures: Patents, Citations, and Citations Q . This suggests that firms with financial experts produce more efficient innovation: financial expert CEOs generate more patents and citations per dollar spent in R&D. Our regression analysis reveals that the sensitivity of

patents to R&D investment is 1.6% higher for financial experts. The sensitivity of citations to R&D spending is up to 2.7% higher for financial experts. Nevertheless, one cannot conclude that this behavior is optimal. Because we find evidence that financial experts invest less in R&D, it might be that they only invest in more successful or profitable projects, rejecting the low profitability ones. This is not

necessarily optimal if they reject low profitability projects that still have positive net present value (NPV). Other alternative interpretations for these results are that financial experts might do less in-house innovation and acquire already patented innovation; or that they only invest in innovation that generates hard evidence of success, i.e., that is more likely to be patented and generate citations.

In summary, we find evidence consistent with financial expert CEOs producing less innovation. We also find their R&D investment to be more efficient in terms of generated output: patents and citations. However, one cannot conclude that this behavior is optimal.

6.2. Advertising

Common wisdom suggests that more financially oriented managers are less willing to spend on marketing-related projects, such as advertising. We empirically test if this is the case. Column 1 of Table 15 shows the results. The dependent variable in column 1 is the log of advertising expenses. We do not find significant evidence that financial expert CEOs spend less on advertising. In fact, when we regress advertising expenses on financial expertise, we find a positive coefficient, but the estimate is not statistically significant.

We then test whether financial experts are more or less efficient in their marketing spending by looking at the sensitivity of sales to advertising expenses. Columns 2 through 4 of Table 15 show the results. As expected, we find a positive sensitivity of future sales to current advertising expenses. Interestingly, we find this sensitivity to be about 31% lower for firms run by financial experts (see column 2) when we estimate the sensitivity using OLS without industry fixed effects. Two possible interpretations follow from this result: Financial experts lack marketing skills which results in sales being less sensitive to advertising, or they tend to be matched to firms/industries that are themselves less sensitive to marketing efforts. Because the previous result is not robust to the inclusion of industry fixed effects, the second interpretation seems the most valid. One cannot argue that financial experts are less effective in marketing activities, but they seem to be matched to industries that are. We find consistent results when we include firm fixed effects in column 4.

In short, we do not find evidence that financial experts invest less in marketing activities, or that they are less efficient in such activities. However, we find evidence consistent with financial experts being matched to industries that have less sensitive revenue to marketing expenses.

6.3. Wages

Lastly, we focus on human-resources (HR) related policies. The financial expertise of the CEO might be associated with HR policies that are less sensitive to the well-being or happiness of the employees. Because well-being and happiness are difficult to measure, we focus on the average wage per worker, which is a more tangible measure of employee satisfaction. Table 15 columns 4 and 5 show the results. The dependent variable in both

columns is the total wage expense divided by the total number of employees in a given year. The coefficient of interest is the financial expert CEO dummy, which would capture any average wage differential between financial and nonfinancial expert CEOs. We do not find any statistically significant differences between firms that are managed by financial expert CEOs and nonfinancial expert CEOs. The point estimate is negative in all specifications, but not statistically different from zero.

Overall, the results in this section do not support the view that financial expert CEOs follow different HR policies.

7. Conclusion

Our analysis of the role of financial expert CEOs in nonfinancial firms indicates that firms headed by financial expert CEOs tend to hold less cash, more debt, and engage more in share repurchases.

Analysis of actions and behavior of these financial expert CEOs at a micro-level yields evidence consistent with a conclusion that financial expert CEOs are more financially sophisticated and are more likely to follow academic theory. We provide evidence that financial experts are less likely to be trapped by the WACC fallacy (using a companywide discount rate instead of a project-specific one). They more dynamically manage their cash and leverage. They are able to increase both cash and debt levels in periods when it is difficult to do so, when default spreads are high, and credit conditions are tight. We find this practice to be advantageous for shareholders. Firms run by financial expert CEOs are less exposed to the riskiness of internal cash flows when they need to invest. They were also more responsive to the dividend and capital gains tax cuts in 2003 and paid out more to shareholders.

We find no evidence that access to better personal networks in the financial sector drive these results. To implement changes in the company, financial expert CEOs also replace incumbent chief financial officers (CFOs) more often. We interpret this result as consistent with the idea that changing management helps to overcome inertia, which makes the implementation of new strategies either feasible or easier.

These results have several implications. We show that previous *work experience* of the CEOs, namely, their financial expertise, affects firms' financial policies. Researchers to date have focused mostly on (partly unobservable) top manager characteristics or on age and education. We also explicitly discuss the concern that endogenous matching between CEOs and firms based on *time-varying characteristics* biases the results. We analyze the matching process itself, and we show some evidence of endogenous CEO-firm matching based on the financial expertise of the CEO and the life cycle of the firm. Results in a battery of additional tests provide evidence that is mostly consistent with a "financial skills view" interpretation of the effects of financial expertise. In addition, we show that financial expertise seems to come at the cost of other skills relevant to the firm, as suggested by the evidence that financial experts produce less innovation.

Concerns about endogenous CEO-firm matching based on time-varying characteristics are likely to be relevant for other research on CEO characteristics as well, a subject for future research.

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